

**Lattice Mismatched Compound Semiconductors
and Devices on Silicon**

by

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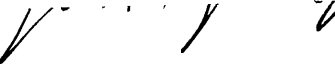
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ABSTRACT

III-V compound semiconductors, due to their superior electron mobility, are promising candidates for n-type metal-oxide-semiconductor field effect transistors (MOSFETs). However, the limited size of III-V substrates and the degradation of III-V MOSFET channel mobility remain two major challenges for III-V MOSFETs. The purpose of this thesis is to solve or partially solve these challenges.

To create large diameter III-V materials, the synthesis of high quality III-V compound semiconductors (lattice-matched to InP) on Si substrates by metal-organic chemical vapor deposition (MOCVD) was studied. Epitaxy of III-V/Si (or Ge) may give rise to antiphase disorders due to the inequivalence of the two face-centered-cubic sublattices of III-V zinc-blende structures. By using a 6° offcut substrate (Ge on insulator) which favors a double-step surface reconstruction upon annealing, antiphase disorders were suppressed and single-domain GaAs on Si was demonstrated. The lattice was then graded from GaAs to InP by compositionally graded $\text{In}_x\text{Ga}_{1-x}\text{As-In}_y\text{Ga}_{1-y}\text{P}$ alloys and $\text{GaAs}_{1-z}\text{Sb}_z$ alloys, which introduce the strain gradually to promote dislocation propagation while suppressing nucleation. It was demonstrated that the phase separation in these pseudobinary alloys could be kinetically suppressed by the low surface diffusivity of adatoms during the non-equilibrium MOCVD process. This allowed us to achieve high quality InP on offcut GaAs. In addition, the dislocation kinetics of $\text{GaAs}_{1-z}\text{Sb}_z$ alloys was studied. The dislocation glide velocity of $\text{GaAs}_{1-z}\text{Sb}_z$ alloys was estimated to be 1 $\mu\text{m/s}$ at 575°C by fitting the experimental data with the dislocation propagation theory.

The channel electron mobility of InP-based $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ quantum-well MOSFETs

was studied by Hall measurements and the dominant scattering mechanisms were discussed. Although invariant for different gate dielectric (Al_2O_3) thicknesses, the mobility turns out to be strongly dependent on the barrier thickness, gate dielectric/barrier interfacial defect states and carrier density. To understand and quantify this dependence, a theoretical model based on internal phonon scattering and interfacial defect coulomb scattering was developed. The Born approximation, random-phase approximation, and two-dimensional limit for carriers were assumed for the coulomb scattering. The results of this model are in good agreement with the experimental data, and the predictions from this model shed light on future MOSFET design and synthesis.

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List of Acronyms and Abbreviations

Arsine.....	AsH ₃
Atomic Force Microscopy.....	AFM
Atomic Layer Deposition.....	ALD
Capacitance-Voltage.....	C-V
Cross-Sectional Transmission Electron Microscopy.....	X-TEM
Density of Interfacial Traps.....	D _{it}
Disilane.....	Si ₂ H ₆
Ge on Insulator.....	GOI
Heterojunction Bipolar Transistor.....	HBT
High Electron Mobility Transistor.....	HEMT
Isopropyl Alcohol.....	IPA
Metal Organic Chemical Vapor Deposition.....	MOCVD
Metal Oxide Semiconductor Field Effect Transistor.....	MOSFET
Molecule Beam Epitaxy.....	MBE
Phosphine.....	PH ₃
Plan-View Transmission Electron Microscopy.....	PV-TEM
Si on Lattice Engineered Substrate.....	SOLES
Standard Cubic Centimeters per Minute.....	SCCM
Threading Dislocation Density.....	TDD
Transmission Electron Microscopy.....	TEM
Trimethylaluminum.....	TMAI
Trimethylantimony.....	TMSb
Trimethylgallium.....	TMGa
Trimethylindium.....	TMIIn
X-Ray Diffraction.....	XRD

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Chapter 1 Introduction

1.1 Motivation

In this thesis we studied the integration of the InP lattice on Si substrates and the InP-based InGaAs metal-oxide-semiconductor field effect transistors (MOSFETs). This research was motivated by the desire to realize III-V/Si MOSFETs.

Why is the InP lattice important? Because of the high electron mobility and unique optical properties, InP and semiconductor alloys that are lattice-matched (or slightly lattice-mismatched) to InP enable various state-of-the-art electronic and optical devices. High speed InGaAs/InP and GaAsSb/InP-based heterojunction bipolar transistors (HBTs) provide important functionality for high frequency electronics applications [1.1, 1.2] and recently, InGaAs/InP based structures have also been extensively investigated for high electron mobility transistors (HEMTs) [1.3], MOSFETs [1.4], and quantum-well MOSFETs [1.5]. InP-based 1.3 and 1.55 μm light sources and detectors are also widely used for low loss fiber-optic communications applications. Why is the integration of InP on Si important? It is because by this monolithic III-V/Si technology, one can realize multi-functionality on one chip and in the mean time lower the total cost. Therefore, integration of the InP lattice on Si is one very important subject.

Why is the InGaAs MOSFET important? MOSFETs are very important devices which are used in microprocessors, memories, etc. Nowadays, Si is used as the channel material in MOSFETs. To keep up with Moore's law, strained Si is

used as the channel material to enhance the mobility and device scaling down is continuously pursued. However, there are several problems for scaling down Si MOSFETs: fabrication difficulty, high off-state current, etc. Therefore, MOSFETs based on III-V semiconductors, carbon nanotubes, graphene ribbons and other novel materials were proposed (Figure 1.1). InGaAs lattice-matched to (or with a larger lattice than) InP has very high electron mobility and is a good candidate for III-V MOSFETs.

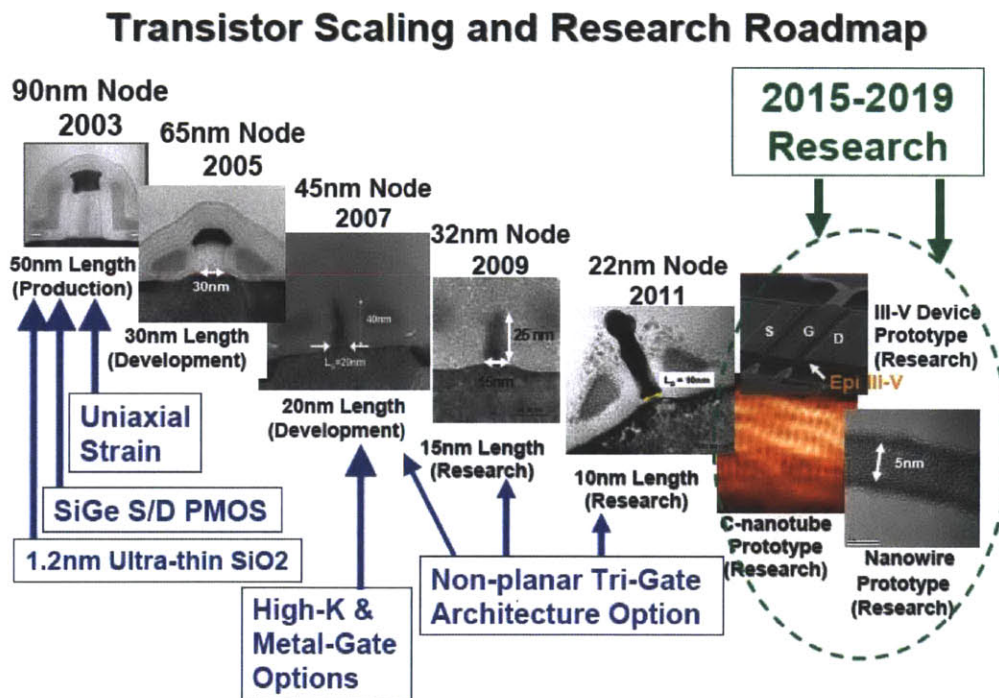


Figure 1.1 Transistor Scaling and Research Roadmap [1.6].

Why is the mobility of III-V MOSFETs important? For III-V MOSFETs, the gate dielectric/III-V interface is not as well established as the gate dielectric/Si interface; therefore the mobility gets degraded due the scattering from the gate dielectric/III-V interface. To study the mobility of MOSFETs will help us

understand the properties of the gate dielectric/III-V interface and provide us with more insight on improving the interface quality. Recently, Radosavljevic *et al.* demonstrated high performance InGaAs quantum-well MOSFET which has a barrier layer in between the channel and oxide/III-V interface to improve the mobility [1.5]. To optimize the design of such quantum-well MOSFET structures, one needs to understand the correlation between the channel mobility and the structural parameters.

Therefore, the integration of an InP lattice on Si and the mobility of InP-based InGaAs MOSFET structures are very important research topics.

1.2 Lattice Engineered Substrates

1.2.1 Background

The progress in lattice-mismatched epitaxy, engineered substrates, and clever process integration has brought the integration of III-V on Si closer to reality. Recently, researchers demonstrated in the DARPA monolithic COSMOS program (by MIT, Paradigm, and Raytheon) that the integration of InP heterojunction bipolar transistors with silicon CMOS is possible, utilizing engineered substrates (silicon on lattice engineered substrate, also called SOLES, Figure 1.2) designed to ease process integration [1.7-1.9]. For the current SOLES wafer as shown in Figure 1.2, it has two Si layers: the top Si device layer and the bottom Si handle wafer. The top Si layer is important because first it is used to build Si devices and secondly it covers the III-V

template layer to make this structure acceptable to a Si fab. The SOLES wafer also has a Ge layer as the III-V growth template for building III-V devices on this layer. However, Ge is lattice matched to GaAs, not lattice matched to InP. Therefore, to make InP-based devices, a metamorphic buffer is necessary to get to the InP lattice. This metamorphic layer will cause two problems: first, the metamorphic layer is typically thick and therefore makes the III-V device layer higher than Si device layer; second, the metamorphic layer will block the heat transfer from the InP device to the substrate. Therefore, for optimal performance, the III-V template layer needs to be InP as shown in Figure 1.3.

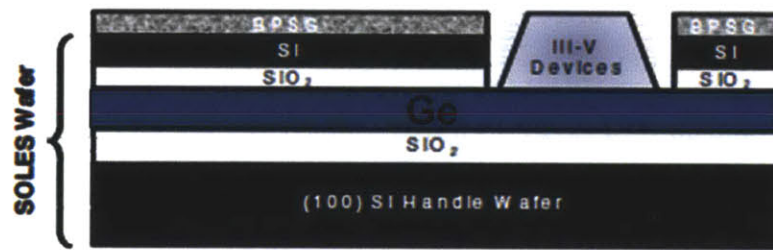


Figure 1.2 Current Si on Lattice Engineered Substrate (SOLES) with Ge as the III-V Template Layer [1.7-1.9].

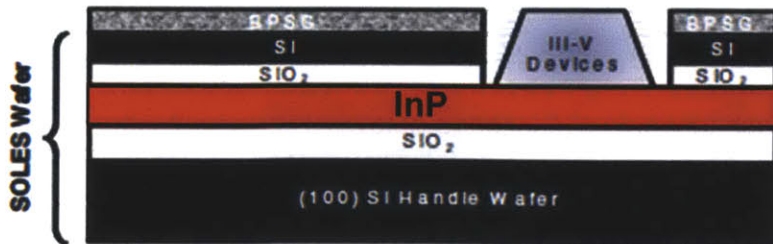


Figure 1.3 Future Si on Lattice Engineered Substrate (SOLES) with InP as the III-V Template Layer.

To create an InP template in SOLES, one must first create a high-quality InP epitaxial layer on Si. After such a metamorphic wafer is created, one can transfer the top thin InP layer to a Si substrate by smart cut and wafer bonding process [1.7, 1.10], forming the base for the SOLES substrate. By transferring another Si thin layer to this base, one will form the final SOLES substrate as show in Figure 1.3. Here I want to emphasize that one cannot directly bond an InP wafer to a Si substrate to form InP on Si because typically wafer bonding needs high temperature, but InP and Si have different coefficients of thermal expansion. Although our group has created SOLES substrates with InP templates by directly depositing InP on silicon substrates, the InP layers are not able to have a threading dislocation density of lower than 10^7 cm^{-2} . In this thesis, we studied how to get to the InP lattice in Figure 1.3 with a threading dislocation density in the 10^6 cm^{-2} range from the Si lattice.

1.2.2 Challenges

There are several technology challenges in transitioning from the Si lattice constant to that of InP (Figure1.4): lattice mismatch, coefficient of thermal expansion mismatch, anti-phase boundaries, and phase separation. We consider each technology challenge below.

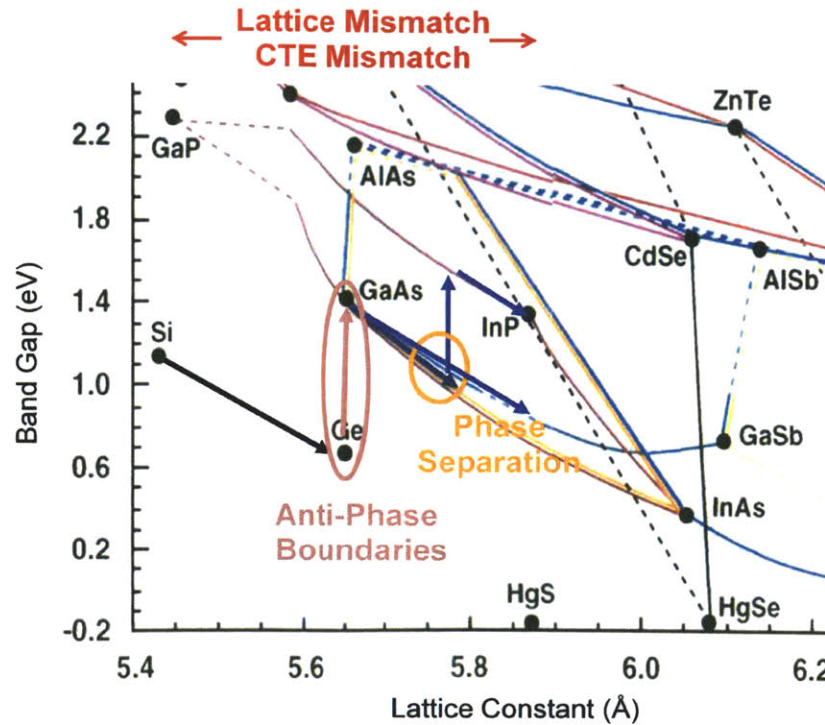


Figure 1.4 Challenges in transitioning from the Si lattice constant to that of InP.

1.2.2.1 Lattice Mismatch

The lattice-mismatch between InP and Si is huge (8.06%). Direct deposition of materials with large lattice-mismatch onto a substrate yields a high dislocation density. This is because a large lattice-mismatch nucleates large quantities of misfit dislocations at the interface and results in a high threading dislocation density (Figure 1.5 (a)). On the other hand, a graded buffer introduces lattice-mismatch gradually and gives a low threading dislocation density. A graded buffer can be imagined as a series of several small lattice-mismatched steps as shown in Figure 1.5 (b). The initial step nucleates dislocations necessary to accommodate the lattice-mismatch between the substrate and the initial step.

These nucleated dislocations will propagate and accommodate the mismatch introduced by the following steps (Figure 1.5 (b)).

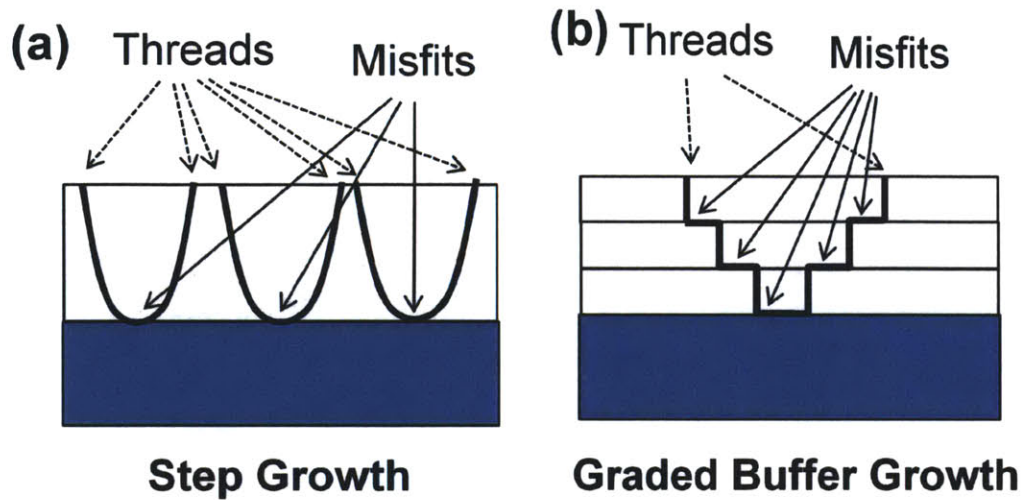


Figure 1.5 Comparison between (a) step growth and (b) graded buffer growth. Graded buffer growth gives lower threading dislocation density.

As indicated in Figure 1.5, graded buffer growth gives a lower threading dislocation density compared to step growth. Therefore graded buffer growth instead of two-step growth is needed to get a low threading dislocation density InP lattice on Si.

1.2.2.2 Thermal Expansion Mismatch

Besides the lattice mismatch, the thermal expansion coefficients of Si and InP are also very different. InP has a thermal expansion coefficient of $4.6 \times 10^{-6} \text{ }^{\circ}\text{C}^{-1}$ [1.11] (most III-V materials have thermal expansion coefficients around $5 \times 10^{-6} \text{ }^{\circ}\text{C}^{-1}$), much larger than that of Si, which is $2.6 \times 10^{-6} \text{ }^{\circ}\text{C}^{-1}$ [1.12]. Therefore, to

prevent the formation of cracks, one needs to limit the thickness of III-V films on Si. Yang *et al.* studied the crack formation for GaAs epi-films on Si and they found that due to the thermal expansion mismatch between GaAs and Si, the critical thickness of a GaAs epi-film for the onset of crack formation is around 5 μm at a growth temperature of 650 $^{\circ}\text{C}$ [1.13]. They also found that if the GaAs was deposited on top of a SiGe graded buffer, the critical thickness is even smaller (around 3 μm at a growth temperature of 650 $^{\circ}\text{C}$) [1.13].

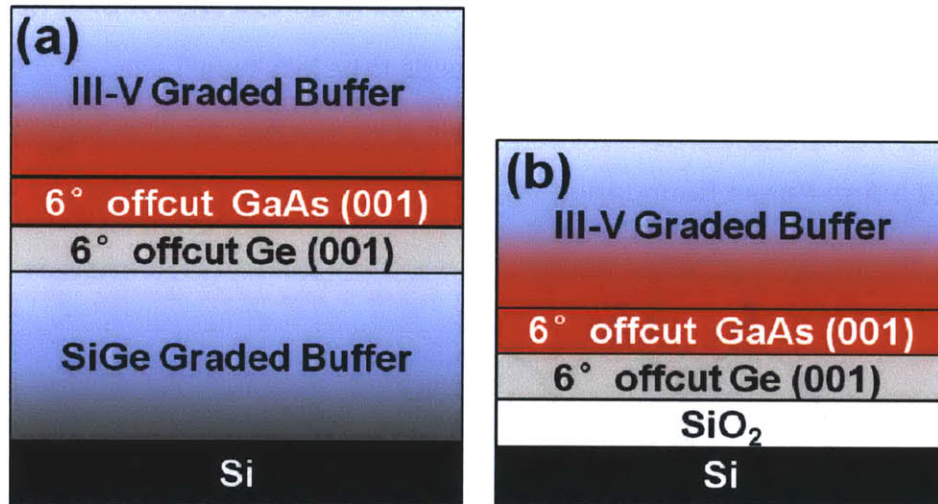


Figure 1.6 Comparison between III-V graded buffers on (a) SiGe graded buffer and (b) Ge on Insulator (GOI) substrates.

As discussed in section 1.2.2.1, due to the lattice mismatch, graded buffer growth instead of step growth is needed to get an InP lattice on Si. However, if we use graded buffer to grade all the way from Si to InP, which means we deposit the III-V graded buffer on top of SiGe graded buffer (Figure 1.6 (a)), the critical thickness for III-V graded buffer will be really small (around 3 μm) [1.13]. Because of the development of smart cut and wafer bonding

technology [1.7, 1.10], nowadays, Ge on Insulator (GOI) templates are commercially available and this can be used to bridge the lattice constant of Si to that of Ge. Therefore a III-V graded buffer can be deposited on GOI to get an InP lattice on Si structure (Figure 1.6 (b)). Using the GOI template only introduces an extra thickness of 100-200 nm instead of a thickness of microns for SiGe graded buffers, which is very important for a thermal budget point of view. Now, the engineering of the lattice constant from Ge to that of InP becomes the key task. Similar to the case of GaAs on Si in reference [1.13], to prevent the formation of cracks, the III-V graded buffer (with growth temperature of 650 °C) on top of GOI needs to be less than 5 μm .

1.2.2.3 Anti-Phase Boundaries

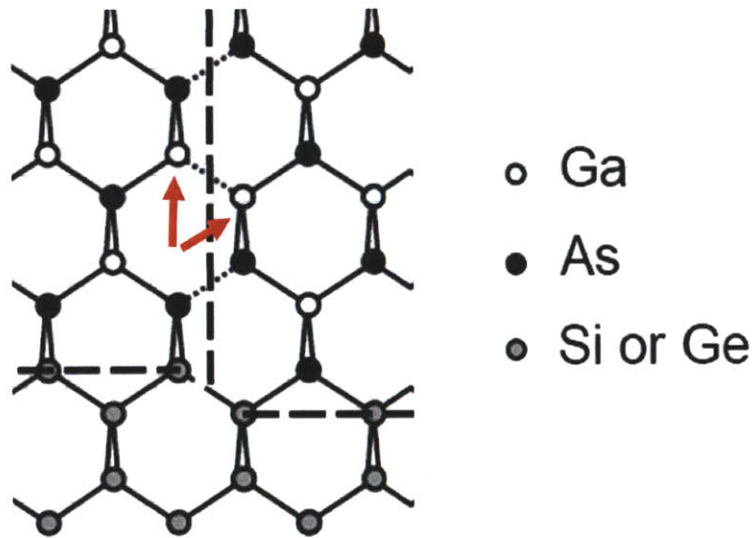


Figure 1.7 Schematic image of antiphase disorder in GaAs/Ge or GaAs/Si growth. The red arrows show Ga-Ga bonding which indicates antiphase disorder.

As shown in Figure 1.4, GaAs is almost lattice matched to Ge, therefore, deposition of GaAs on Ge does not introduce a large amount of dislocations. However, deposition of GaAs on Ge, a compound semiconductor on an elemental semiconductor is non-trivial. Due to the inequivalence of its two face-centered-cubic sublattices, the zincblende GaAs has lower symmetry than the diamond cubic Ge. Therefore, when GaAs is nucleated on a Ge surface, two GaAs sublattice orientations are both possible, which leads to two distinct domains separated by anti-phase boundaries as shown Figure 1.7. Anti-phase boundaries consist of planes of wrong nearest-neighbor bonds as indicated by the red arrows (Figure 1.7). By choosing offcut substrates and proper annealing conditions, our group demonstrated the growth of GaAs on Ge without anti-phase boundaries by metal-organic chemical vapor deposition (MOCVD) [1.14]. The offcut provides the step structures on the Ge surface and proper annealing conditions make a Ge surface consisting of mostly double steps, which are essential to suppress anti-phase disorder [1.14].

By using similar growth conditions of reference [1.14], we successfully demonstrated GaAs on a 6° offcut GOI substrate (Figure 1.8). This structure bridges the lattice constant of Si to that of GaAs with a very low threading dislocation density, no anti-phase disorder, and a thin epi-film. Therefore, our focus mainly becomes engineering of the lattice constant from that of GaAs to InP. Specifically, our objective is to establish the InP lattice constant on 6° offcut (001) GaAs substrates, while maintaining a low threading dislocation density ($\sim 10^6 \text{ cm}^{-2}$) and a surface morphology suitable for eventual device processing. The specification of the 6° miscut is important because the

suppression and elimination of antiphase disorder during the growth of III-V compounds on group IV semiconductors requires the step structure provided by the miscut [1.14]. Therefore, eventual integration of the InP lattice constant with Si substrates requires the understanding of how to integrate the InP lattice on 6° offcut GaAs.

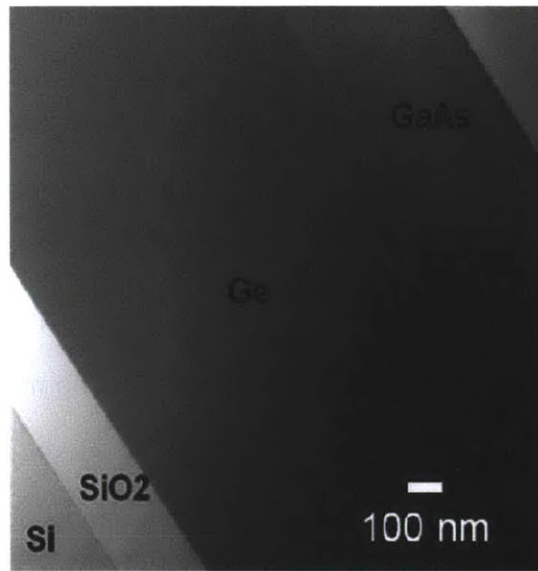


Figure 1.8 Cross-sectional transmission electron microscopy (X-TEM) of GaAs on a 6° offcut GOI substrate. The Ge layer in the GOI substrate can be further thinned by a chemical etch prior to the growth.

1.2.2.4 Phase Separation

To integrate the InP lattice on offcut GaAs, we need to use III-V alloy graded buffers to accommodate the lattice-mismatch. In III-V alloy systems, one very common defect is phase separation, which is known as the composition modulation in the alloy film. Phase separation usually pins dislocations and

greatly increases the threading dislocation density. Figure 1.9 shows phase separation in the $\text{In}_{0.3}\text{Ga}_{0.7}\text{As}$ graded buffer on a 6° offcut GaAs (001) substrate grown at 700°C by MOCVD. The phase separation here in the InGaAs graded buffer is due to the thermodynamic and kinetic effects of the surface during epitaxy instead of the thermodynamic properties of the bulk material. According to the phase diagram of the InAs-GaAs system [1.15] as shown in Figure 1.10, $\text{In}_{0.3}\text{Ga}_{0.7}\text{As}$ should not phase separate at 700°C .

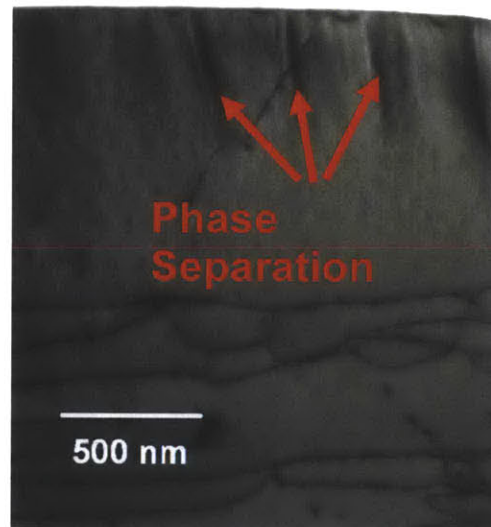


Figure 1.9 XTEM of the $\text{In}_{0.3}\text{Ga}_{0.7}\text{As}$ graded buffer on a 6° offcut GaAs (001) substrate grown at 700°C by MOCVD. The dark-bright contrast indicates the onset of phase separation.

As discussed above, the phase separation in the InGaAs buffer is surface driven during epitaxy. This kind of phase separation can be effectively suppressed by lowering the diffusion length of the adatoms. The diffusion length of adatoms is determined by their surface diffusivity and diffusion time.

Therefore, to suppress phase separation, one can either reduce the adatom surface diffusivity (e.g., lower the growth temperature or choose low diffusivity atoms) or reduce the diffusion time (e.g., increase growth rate). In this thesis, we lowered the growth temperature of the InGaAs graded buffer to 450°C to suppress its phase separation. We also chose the GaAsSb alloy as the graded buffer to get to the InP lattice, where phase separation was suppressed by the low surface diffusivity of Sb over the graded region.

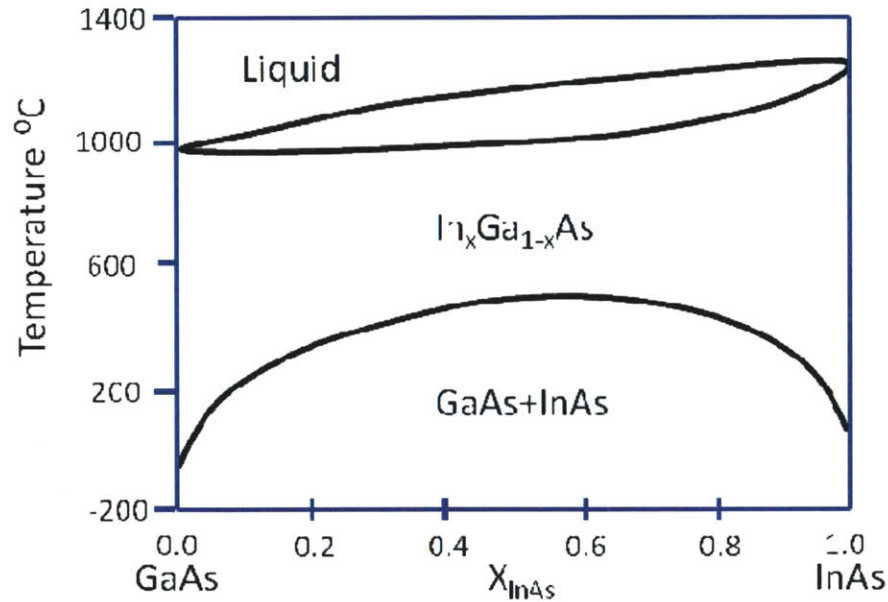


Figure 1.10 Pseudobinary phase diagram for the InAs-GaAs system [1.15].

1.2.3 Our Approach

In this thesis, we studied two approaches to engineer the InP lattice constant from that of GaAs. The first approach consisted of tandem graded layers of InGaAs and InGaP with compositional grading of the In concentration (Figure

1.11 (a)). This tandem approach was found to be necessary because phase separation in the InGaAs alloys leads to surface roughening and a high threading dislocation density when grading to lattice constants that are greater than that of $\text{In}_{0.30}\text{Ga}_{0.70}\text{As}$. An $\text{In}_x\text{Ga}_{1-x}\text{As}$ graded buffer was grown at 700°C for low In concentration ($X_{\text{In}}=0-0.10$), and then the growth temperature was decreased to 450°C for high In concentration ($X_{\text{In}}=0.10-0.30$) to suppress the phase separation. The growth temperature was then increased to 650°C and the graded $\text{In}_y\text{Ga}_{1-y}\text{P}$ system was implemented to continue grading the lattice constant from $\text{In}_{0.30}\text{Ga}_{0.70}\text{As}$ to InP. The second approach used GaAsSb alloys with a compositional grading of the Sb concentration (Figure 1.11 (b)). Graded mixed-anion GaAsSb alloys grown at 575°C did not exhibit phase separation, resulting in high quality InP lattice constant films on GaAs without the need to transition to another material system for compositional grading.

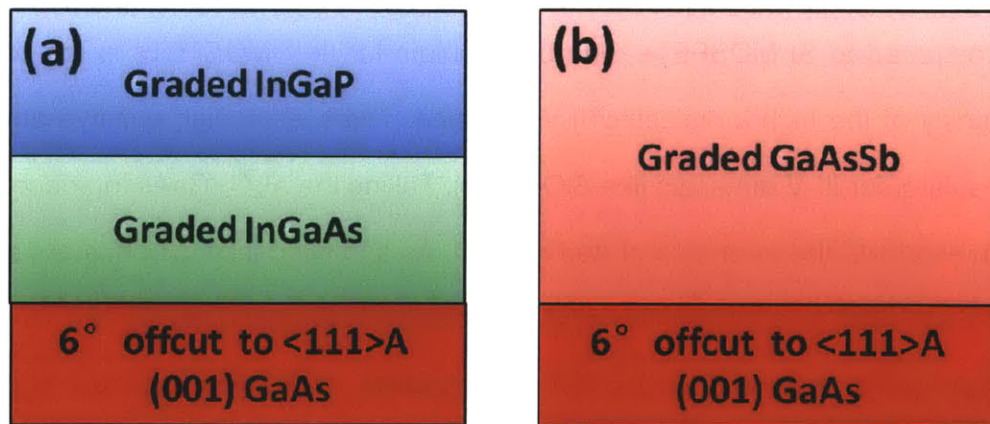


Figure 1.11 Transition from the GaAs lattice constant to that of InP by compositionally-graded (a) InGaAs-InGaP alloys and (b) GaAsSb alloys.

1.3 Mobility of III-V MOSFETs

1.3.1 Background

In the past 40 years, Si MOSFET technology continuously followed Moore's law by scaling down. However, when the Si MOSFET dimension goes below 10 nm, short channel effects, such as high off-state current, become very significant. On the other hand, MOSFET technology under 10 nm is approaching the photo lithography limit, which brings fabrication difficulties. As Si MOSFET technology is approaching its function and fabrication limits, non-Si MOSFET technology based on III-V semiconductors, carbon nanotubes, graphene ribbons and other novel materials starts to gain more attention in both academia and industry. Due to their high electron mobility, III-V compound semiconductors, especially $\text{In}_x\text{Ga}_{1-x}\text{As}$ alloys, become promising candidates for high-performance n-MOSFETs.

Compared to Si MOSFETs, one big problem for III-V MOSFETs is the poor quality of the high-k dielectric/III-V interface. There is no high quality natural insulator for III-V materials like SiO_2 for Si. Taking the $\text{Al}_2\text{O}_3/\text{GaAs}$ interface as an example, the existence of As_xO , Ga_yO , As and Ga dangling bonds, As and Ga anti-sites give a high interfacial trap density in the band gap [1.16]. To get high quality high-k dielectric/III-V, researchers have tried different high-k dielectric materials, including Al_2O_3 [1.17, 1.18], Ga_2O_3 [1.19], HfO_2 [1.20], etc. Researchers also have performed surface preparation to reduce the interfacial defect density and improve semiconductor device performance by using an ammonium sulphide ($(\text{NH}_4)_2\text{S}$) wet etch [1.21] or self-cleaning under

Arsine (AsH₃) overpressure [1.22] or pretreatment with trimethylgallium (TMGa) [1.16] prior to the high-k dielectric deposition. Different growth schemes [1.17, 1.23] and the insertion of a Si interlayer [1.20, 1.24] were also explored to improve interface quality. After more than 35 years of study of high-k dielectric/III-V interfaces, researchers have made progress in reducing the Fermi level pinning effect for high In concentration InGaAs MOSFETs [1.24]. However, the typical interfacial trap density for high-k dielectric/III-V interfaces in the band gap is still in the range of 10^{12} - 10^{15} cm⁻²eV⁻¹ [1.25] and the identification of the bonding configurations that cause the defects still remains challenging [1.26].

Compared to Si, one big advantage for III-V compound semiconductor, especially In_xGa_{1-x}As, is the high electron mobility. To keep the mobility advantage of In_xGa_{1-x}As, a lot of work was done on In_xGa_{1-x}As high electron mobility transistors (HEMTs) which incorporate an undoped quantum-well channel and use Schottky gate contacts to modulate the charge density within the channel [1.27]. However, the Schottky gate contact leads to large gate leakage currents and prevents the large scale integration of these transistors. On the other hand, In_xGa_{1-x}As metal oxide semiconductor field effect transistors (MOSFETs) can exhibit very low gate leakage because of the chosen gate dielectric, but the need for doping in the channel and the direct contact between the channel and the gate dielectric leads to carrier scattering that reduces carrier mobility and overall device speed. M. Radosavljevic *et al.* established an integrated high-κ gate stack with an In_xGa_{1-x}As HEMT structure and demonstrated a high performance In_xGa_{1-x}As quantum-well

MOSFET [1.5]. P. Nagaiah *et al.* reported on their analysis of channel mobility in $\text{In}_{0.77}\text{Ga}_{0.23}\text{As}$ quantum-well MOSFET structures [1.28].

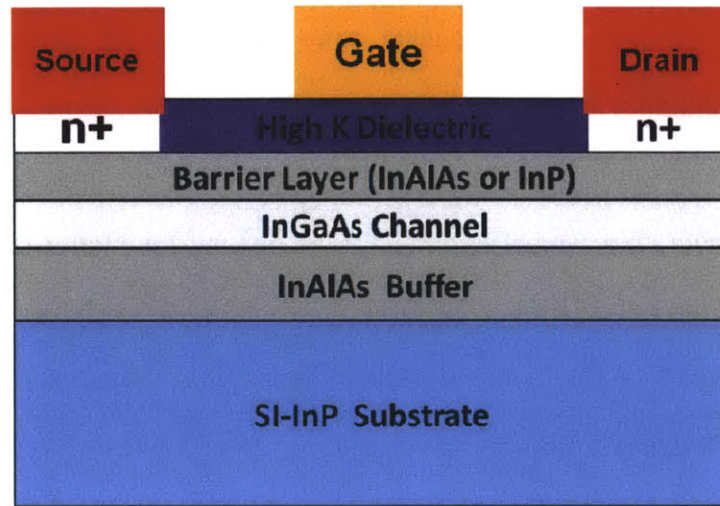


Figure 1.12 Schematic structure of InGaAs quantum-well MOSFETs.

For such a quantum well MOSFET structure (Figure 1.12), the gate leakage current is small because of the gate dielectric; the channel carrier mobility is high because of both the undoped channel and the barrier layers between the channel and the gate dielectric. While the barrier layer(s) is necessary to maintain high carrier mobility in the channel, it also increases the equivalent oxide thickness which is undesirable. Therefore, the optimization of such a quantum-well structure is a very important subject.

1.3.2 Scattering Mechanisms

Similar to the Si MOSFET case [1.29], there are six major scattering mechanisms as shown in Figure 1.13 that may affect the carrier mobility of a

quantum-well MOSFET (with a thick high-k dielectric material and no gate metal): semiconductor (internal) phonon scattering, gate dielectric (remote) phonon scattering [1.30, 1.31], gate dielectric/barrier layer interfacial roughness scattering, coulomb scattering from ionized dopants if there is a doping layer, coulomb scattering from charges trapped in the gate dielectric, and coulomb scattering from the gate dielectric/barrier layer interfacial charge.

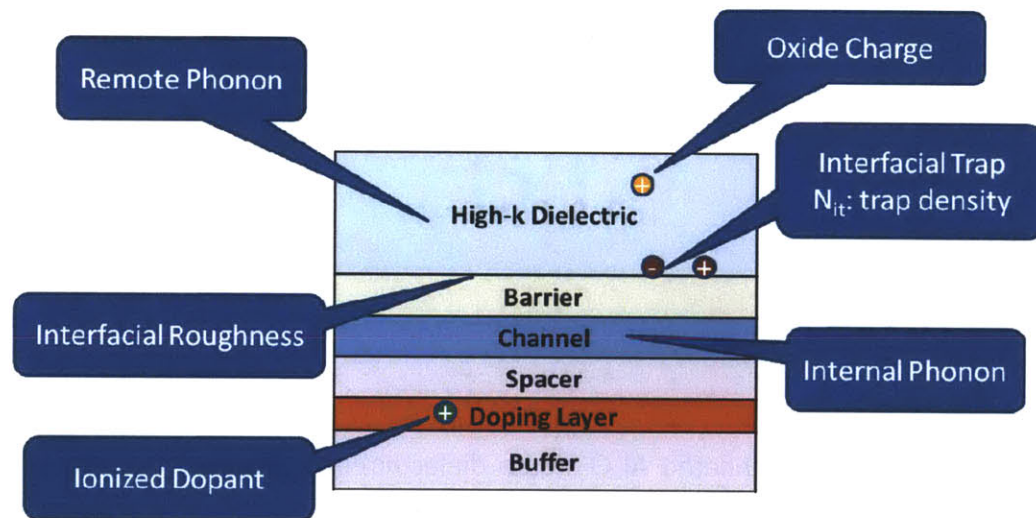


Figure 1.13 Possible scattering sources in quantum-well MOSFET structures.

One important thing to point out here is that charged interfacial traps can reduce the mobility but do not necessarily shift the flat-band voltage of MOSFETs [1.29]. This can be explained if there are both negative and positive charges at the gate dielectric/barrier layer interface, but the net charge is neutral. In this case, the carriers still get scattered by the local disturbance of these negative and positive charges but the flat-band voltage does not shift due to the neutral net charge [1.29].

1.3.3 Our approach

ALD or CVD Al_2O_3 t_{oxide}	
InGaAs Cap	} t_{barrier}
InAlAs Barrier	
InGaAs Channel	15 nm
InAlAs Spacer	15 nm
Si δ doping	
InAlAs Buffer	30 nm
InGaAs Interlayer	0.3 nm
InP Epi	5 nm
InP SI Substrate	

Figure 1.14 Schematic of quantum-well MOSFET structures fabricated in this thesis.

In this thesis, we fabricated Al_2O_3 (gate dielectric)/ $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ - $\text{In}_{0.52}\text{Al}_{0.48}\text{As}$ (barrier)/ $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ (channel) structures as shown in Fig. 1.14 to study the effect of different scattering mechanisms on the carrier mobility. For such a buried channel MOSFET with no gate bias, it is reasonable to assume that the Al_2O_3 phonon scattering [1.32] and interfacial roughness scattering would be negligible. In addition, for our structure the effects of ionized impurity scattering due to the Si δ doping layer could also be considered negligible since the spacer layer was 15 nm thick.

Therefore, in this work, we only consider three scattering processes for our experiments and model: semiconductor phonon scattering, coulomb

scattering from charges trapped in the Al_2O_3 gate dielectric, and interfacial charge density at the $\text{Al}_2\text{O}_3/\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ interface. Hall measurements were used to examine the mobility and a model was developed to fit the experimental data.

For our experiments we first changed the Al_2O_3 thickness (t_{oxide}) to determine whether scattering from charges trapped in the oxide is a key factor in determining the channel carrier mobility. We then fixed the oxide thickness and varied the following three factors to study the effect of coulomb scattering from the interfacial charges: the separation between the interfacial charges and the carriers, the density of the interfacial charges, and the density of the carriers. Specifically, we varied the barrier thickness (t_{barrier}) to change the separation between the interfacial charges and the carriers, varied the Al_2O_3 deposition methods to change the density of the interfacial charges, applied a gate voltage to change the carrier density, and then studied the mobility versus t_{barrier} , the mobility versus different Al_2O_3 deposition methods, and the mobility versus gate voltage.

Based upon the experimental data we developed a model that could be used to sample MOSFET designs that are limited by phonon scattering and gate dielectric/barrier layer interfacial charge density.

1.4 Future Monolithic MOSFETs

Figure 1.15 gives the schematic structures of future monolithic MOSFETs. In

such monolithic MOSFET structures, the n-type FETs would be based on III-V compound semiconductors most likely InGaAs and the p-type FETs would be based on Ge or other III-V materials; however, the n-type FETs and p-type FETs both need to be built on a common Si substrate. Motivated by the desire to realize this kind of monolithic MOSFETs, we studied the integration of the InP lattice on Si substrates and the effect of a high-k/III-V gate stack on the III-V MOSFET carrier mobility as shown in Figure 1.15.

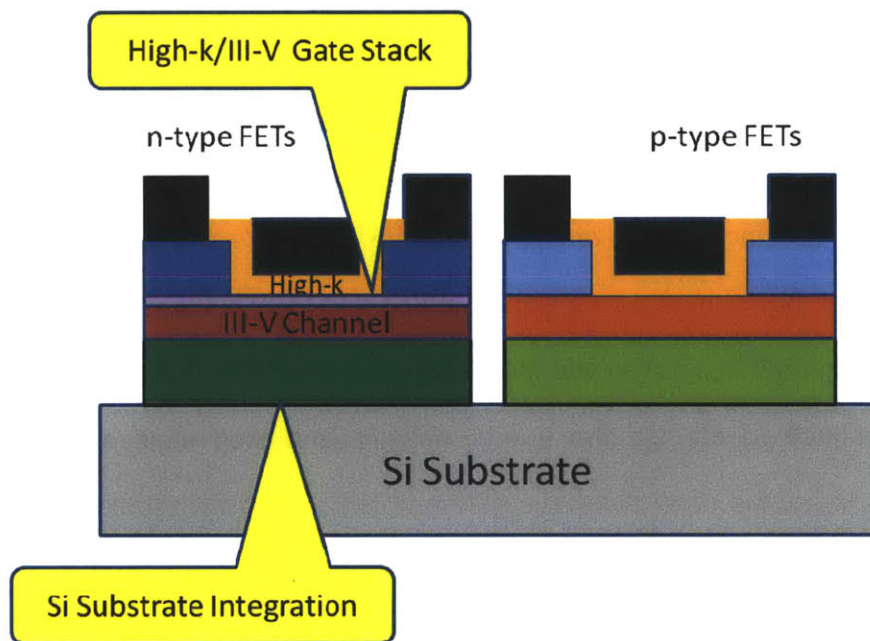


Figure 1.15 Schematic of future monolithic MOSFET structures.

1.5 Organization of the Thesis

This thesis includes five chapters. Chapter 1 is the introduction, which gives the motivation and challenges for this thesis work. Chapter 2 introduces the methods that we used for the materials growth and characterization. Chapter

3 presents the results of integration of the InP lattice on offcut GaAs by using compositionally-graded InGaAs-InGaP alloys as well as compositionally-graded GaAsSb alloys. Chapter 4 presents the experimental and modeling results on the mobility of InGaAs quantum-well MOSFET structures. Chapter 5 summarizes the work of this thesis and offers suggestions for future work.

Chapter 2 Materials Growth and Characterization

In this chapter, we first introduce the growth of III-V materials and high-k dielectric materials in our MOCVD system; then we introduce the techniques used to characterize the materials; finally, we introduce the electrical measurements that we used in this thesis.

2.1 Materials Growth

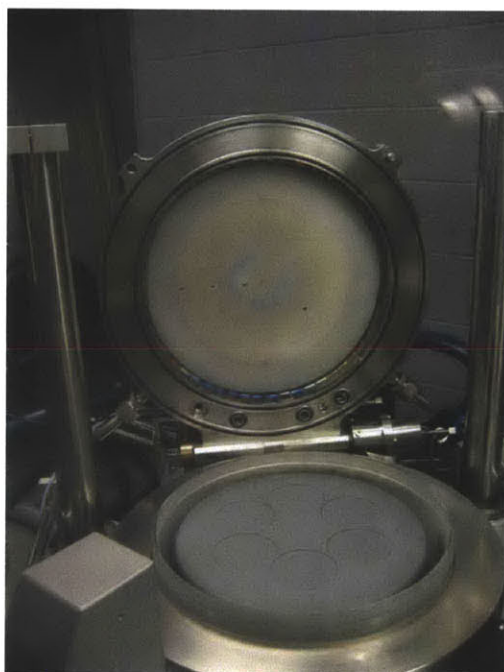


Figure 2.1 Photograph of the MOCVD chamber that was used to grow the samples reported in this thesis.

Deposition of the structures in this thesis project was conducted using a Thomas Swan/AIXTRON low pressure MOCVD reactor with a close-coupled showerhead as shown in Figure 2.1. The III-V compound epitaxy was conducted in the chemical vapor deposition (CVD) mode and Al_2O_3 was deposited *in-situ* in either the atomic layer deposition (ALD) mode or the CVD

mode. This MOCVD machine has several advantages [2.1]. First, it is computer controlled and therefore provides precise control over the gas flow rate, gas flow time, gas switching, pressure in the lines, etc. Second, it uses a close-coupled showerhead which has two sets of small tubes with one set for hydrides and the other for metal-organic precursors. The showerhead is designed to be very close to the substrates but not too close, to make sure that the precursors will mix evenly at 1 mm above the substrates. This design effectively prevents the vapor phase reaction. Third, substrates are placed on top of a rotating susceptor (wafer holder). The rotation of the susceptor improves the uniformity. Fourth, the susceptor is heated by a three-zone graphite heater, and the three-zone design enables adjustment of the temperature profile to provide temperature uniformity over the susceptor diameter. Because of the above advantages, the films grown in this MOCVD reactor show good qualities, e.g., particle-free surfaces (no vapor phase reaction) and good uniformity.

2.1.1 MOCVD of III-V Materials

CVD is a chemical process which is widely used in the semiconductor industry to produce high purity thin films. In a typical CVD process, the substrate is exposed to precursors, which decompose and react on the substrate surface to form solid films. Since lots of precursors are metal-organic, the CVD process is also called metal-organic chemical vapor deposition (MOCVD). An MOCVD process typically includes five steps as shown in Figure 2.2 [2.2]: (1) diffusion of precursors to the substrate through the boundary layer; (2)

absorption of precursors to the substrate surface; (3) surface reaction; (4) desorption of by-products of the reaction; and (5) diffusion of by-products through the boundary layer. The five-step MOCVD process depends on several process parameters and, on the other hand, determines the final properties of the deposited films as discussed below.

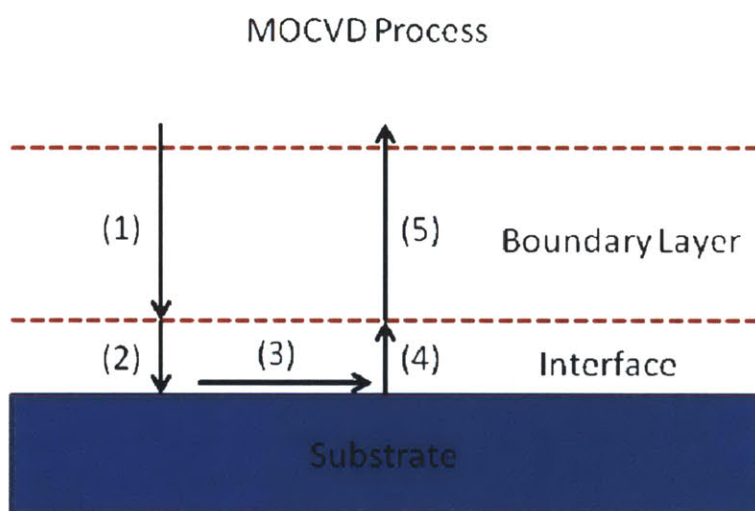


Figure 2.2 Schematic illustration of a CVD process [2.2].

In the MOCVD process of III-V materials, there are three important properties of the deposited films: film thickness, defect density, and composition (for ternary or quaternary materials). These three properties depend on five process parameters, which include growth temperature, chamber pressure, carrier gas flow rate, precursor flow rate, and growth time. Chamber pressure and carrier gas flow determine the boundary layer thickness and in consequence affect the growth rate and film composition. We typically keep the chamber pressure at 100 Torr and the carrier gas flow rate at 20,000 standard cubic centimeters per minute (SCCM). The precursor flow rate and

growth time are set to get targeted film composition and thickness. The precursor flow rate also determines the V/III ratio, which needs to be controlled to give a good stoichiometric film and a droplet-free surface. The growth temperature is a very important parameter here and needs to be carefully controlled due to three reasons that will be discussed in detail below.

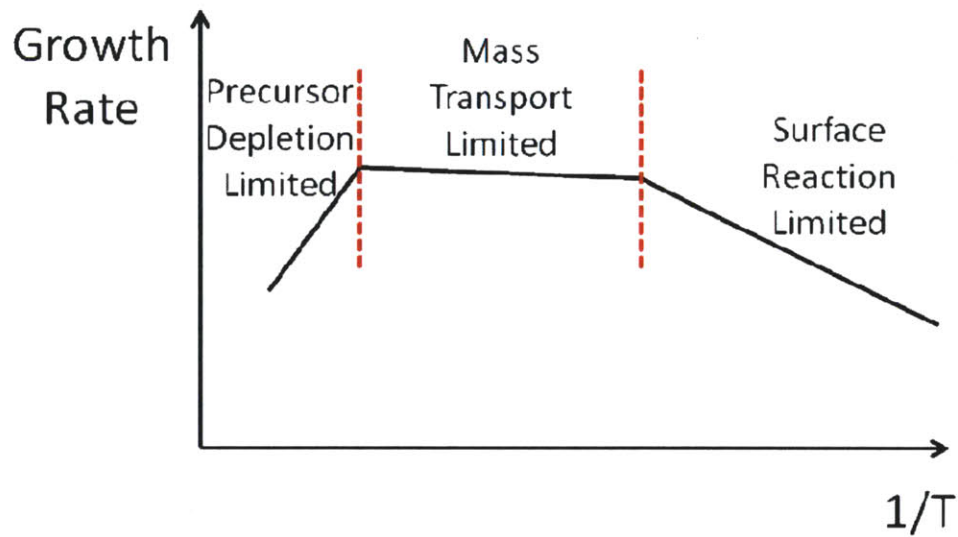


Figure 2.3 Dependence of growth rate on temperature for the MOCVD [2.3].

First, it determines which regime the growth will fall into [2.3]. As shown in Figure 2.2, when the temperature is very low, the growth is in the surface reaction limited regime where the growth rate increases with increasing temperature but is independent of the precursor flow rate; when the temperature is in the medium range (typically 400°C-800°C for III-V arsenides or phosphides), the growth will be in the mass transport limited regime, where the growth rate is dominated by the precursor flow rate, not the growth temperature; when the temperature is very high, the growth is in the

desorption limited regime and the growth rate actually decreases with increasing growth temperature. We want to grow the III-V materials in the mass transport limited regime, because in this case we can control the growth rate and composition by controlling the precursor flow rate. The mass transport limited regime is also good for thickness uniformity control because the growth rate does not strongly depend on growth temperature. Therefore, even if there is temperature fluctuation over the wafer, the growth rate will still remain uniform. Another reason why the growth temperature is an important parameter is that it affects the dislocation gliding rate and in consequence, determines the threading dislocation density of the graded buffer. Typically, increasing the growth temperature increases the dislocation gliding rate and gives low threading dislocation density. Therefore, we want to keep the growth temperature high except in one case, which will be discussed next. Lastly, temperature is important in determining the onset of the phase separation. Phase separation usually pins dislocations and greatly increases the threading dislocation density in a graded buffer system. As discussed in section 1.2.2.4, in an MOCVD process, phase separation can be effectively suppressed by lowering the growth temperature to reduce the adatom surface diffusivity. Therefore, to prevent phase separation, one needs to grow films at low temperature. The tradeoff between the dislocation gliding rate and phase separation typically determines our growth temperature. For example, for a film which does not phase separate, we typically grow it at high temperature, while for a film which is prone to phase separate, we grow it at low temperature.

Detailed MOCVD growth parameters for the III-V materials in this thesis will be discussed in the corresponding chapters.

2.1.2 ALD and CVD of High-k Materials

High-k materials, which are candidates for gate dielectrics, have attracted increasing attention in recent years. With a high-k material, a thicker gate dielectric can be made for the required capacitance density, while the gate leakage current can be reduced through the structure. In 2007, Intel reported that they deposited high-k gate dielectric for their 45 nm CMOS technology [2.4]. High-k materials can be deposited in either the ALD or CVD mode, which may give a different interface chemistry and interfacial defect density.

ALD is similar to CVD, except in an ALD process, the precursors flow into the chamber separately instead of simultaneously. After each precursor pulse, a purge gas flows into the chamber to remove excess precursor, which prevents a possible CVD process. In this case, the precursor only reacts with approximately one monolayer of the previous precursor and this makes ALD a self-limiting process. This self-limiting process allows ALD to control the film thickness by the number of precursor cycles rather than by the deposition time. One complete cycle results in the deposition of one monolayer of the compound.

An ALD process typically includes four steps as shown in Figure 2.4 [2.5]: (1) A self-terminating reaction of the 1st precursor; (2) A purge to remove the non-

reacted reactants and the reaction by-products; (3) A self-terminating reaction of the 2nd precursor; (4) A purge to remove the non-reacted reactants and the reaction by-products.

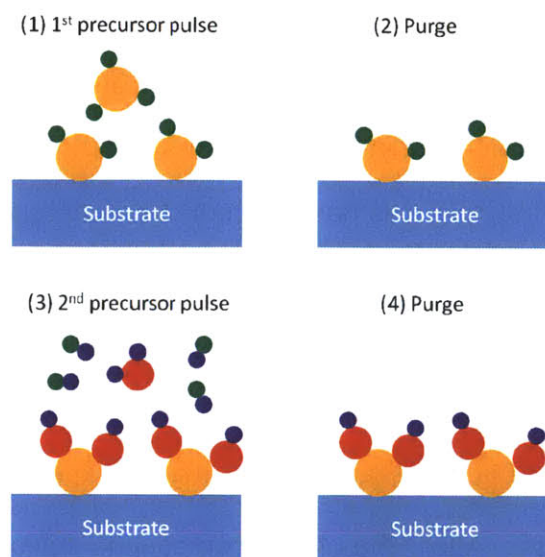


Figure 2.4 Schematic representation of an ALD process [2.5].

In this work, ALD of Al_2O_3 was performed in an MOCVD system by the method developed by Chengwei Cheng in our group [2.6, 2.7]. Trimethylaluminum (TMAI) and isopropyl alcohol (IPA) were used as the precursors. The chamber pressure was set at 50 Torr. High-purity N_2 was used as the carrier gas and the purge gas. The growth temperature was set at 370°C . The pulse time for TMAI (with a partial pressure of 0.03 Torr) and IPA (with a partial pressure of 0.20 Torr) was kept at 4 sec, and the N_2 purge time in between TMAI and IPA pulses was also kept at 4 sec.

The high-k material Al_2O_3 can also be deposited in the CVD mode where two precursors, TMAI and IPA, flow into the chamber simultaneously. In the CVD process of Al_2O_3 , TMA and IPA first react to form the intermediate precursor, aluminum isopropoxide, and then the aluminum isopropoxide thermally decomposes at the growth surface to form Al_2O_3 [2.8]. In our CVD of Al_2O_3 , the chamber pressure was set at 50 Torr and the growth temperature was set at 500°C. TMAI and IPA partial pressures were kept at 0.01 and 0.15 Torr, respectively. This CVD process has different chemical reactions compared to that of the ALD process and should give different high-k/III-V interface properties [2.8].

One thing that I want to emphasize here is that our ALD or CVD of Al_2O_3 was an in-situ process. After the III-V compound epitaxy sequence, the temperature was ramped down to a growth temperature of Al_2O_3 under the group V overpressure and the deposition of Al_2O_3 was then conducted. This ensured that the III-V compound surfaces were never exposed to air prior to deposition of the gate dielectric.

2.2 Structural Characterization

2.2.1 Transmission Electron Microscopy (TEM)

TEM is a microscopy technique which is used to image thin films, nano structures, etc. An image is observed by passing a beam of high energy electrons (eg. 200KeV) through a thin specimen and focusing the electron

beam onto a fluorescent screen or CCD camera. The image is formed because of the interaction of the electrons and the specimen. Due to the small de Broglie wavelength of electrons, the imaging resolution for a TEM can be on the order of angstroms.

TEM is commonly used in characterizing the thickness and defects of semiconductor thin films. Since different atoms have different electron configurations, they interact with the electron beam differently and this results in the different brightness of the transmitted beam. This brightness difference is called “Z-contrast”, where Z is the atomic number. Therefore, different materials will lead to “Z-contrast” and this is often used to characterize the thickness of films. The TEM image can also show contrast of different lattice constants, which can be used for characterization of defects. Defects, like dislocations, typically distort the local lattice and this can show up in a TEM image. Taking dislocation characterization for example, when the diffraction vector $\mathbf{g}$ and the Burgers vector $\mathbf{b}$ meet the criteria $\mathbf{g} \cdot \mathbf{b} \neq 0$, dislocations will be visible [2.9-2.11], where the diffraction vector $\mathbf{g}$ (labeled by Miller indices, e.g., $\mathbf{g} = \langle 220 \rangle$) denotes the direction of the electron beam relative to the specimen and the Burgers vector $\mathbf{b}$ represents the direction and magnitude of the lattice distortion from dislocations in the specimen.

In this thesis, cross-sectional TEM (X-TEM) and plan-view TEM (PV-TEM) were both carried out to characterize thin films, where X-TEM allows us to examine the thicknesses and microstructural characteristics of the films and PV-TEM allows us to examine the threading dislocations and surface

morphology of the films. 30 PV-TEM images ($670 \mu\text{m}^2$ of area) were typically used to determine the threading dislocation density (TDD), which gave the TDD an uncertainty range (95% confidence interval) of less than 1/3 of the total TDD. For X-TEM preparation, two small pieces of samples were stuck together face-to-face like a sandwich and the sandwich was then hand polished from its lateral sides to around $10 \mu\text{m}$. The polished sandwich was then stuck onto a TEM grid and ion milled by using an ion milling machine. For PVTEM preparation, one small piece of sample was hand polished from its back side to around $10 \mu\text{m}$. The polished sample was then stuck onto a TEM grid and ion milled only from the back side of the sample.

In this thesis, both X-TEM and PV-TEM were imaged in either a JEOL 200CX or JEOL 2011 microscope operated at 200 KV. Two main diffraction conditions were used in this work. The first condition is the on-pole diffraction condition, where Z-contrast is enhanced and the layer thickness can be obtained. The second one is $\mathbf{g} = \langle 220 \rangle$, where dislocation contrast is enhanced and dislocation images can be obtained.

2.2.2 Atomic Force Microscopy (AFM)

Atomic force microscopy is a microscopy technique which is used to characterize the surface morphology of samples with a resolution on the order of fractions of a nanometer. The AFM has a cantilever with a sharp tip and when the tip is brought near a sample surface, forces between the tip and the sample lead to a deflection of the cantilever, which can be measured by laser

interferometry. The deflection of the cantilever can then be translated into a topographic image of the sample surface.

AFM can be operated in several modes, depending on the application. In this thesis, AFM was used in the tapping-mode. In the tapping-mode, the AFM tip does not contact the sample surface and the cantilever is continuously vibrated at a specific frequency. Since the cantilever frequency can be affected by the van der Waals force between the tip and the sample, a feedback loop system is used to maintain a constant vibration frequency by adjusting the distance between the tip and the sample, which can then be used to construct the surface morphology of the sample.

In this thesis, AFM was conducted with a Veeco Nanoscope IV AFM to obtain the surface morphology and surface roughness of samples.

2.2.3 X-Ray Diffraction (XRD)

XRD is a technique which is used to examine the atomic structure of materials. It is based on the elastic scattering of X-rays, primarily from the electrons of the atoms in the material, and the interference of these scattered X-rays. In this thesis, we used XRD to get the lattice constants of III-V alloys and then to calculate the composition of these III-V alloys. Typically, a crystal structure can be viewed as different sets of planes of atoms. For a given set of atomic planes, the condition (known as the Bragg condition) for a diffraction peak is:

$$2d\sin(\theta_B) = n\lambda,$$

where d is the inter-plane distance, θ_B is the Bragg angle, n is an integer, and λ is the wavelength of the x-ray. For a set of planes (xyz) in a cubic crystal, d is given by

$$d = \frac{a}{\sqrt{x^2 + y^2 + z^2}}$$

where a is the lattice-constant. Typically, the in-plane and out-of-plane lattice constant of a film with a zincblende lattice can be determined by measuring the Bragg angles for the symmetric (004) and asymmetric (224) sets of planes [2.12]. For more details about the XRD technique, please read McGill's thesis [2.13].

In this thesis, XRD was conducted with either a Philips PANalytical X'pert Pro XRD system or Bruker D8 XRD system operated at 40KV.

2.3 Electrical Measurements

2.3.1 Capacitance-Voltage (C-V) Characteristics

As discussed in section 1.3.1, one big problem for III-V MOSFET is the high interfacial trap density at the gate dielectric/III-V interface. These interfacial defects can be characterized by C-V measurement of a MOS capacitor structure. The density of interfacial traps (D_{it}) can also be obtained from the conductance-frequency method. In the C-V measurement, a voltage is applied on the MOS capacitor structure as shown in Figure 2.5 (a) and the capacitance and conductance is then measured. The voltage V_G applied

across the MOS capacitor is a superposition of a DC bias (V_{DC}) and a very small AC bias (V_{AC}) as shown in Figure 2.5 (b). The DC bias is used to tune the Fermi level and band-bending of the semiconductor and the AC bias is used to get the capacitance ($C=dQ/dV_{AC}$) and conductance ($G=dl/dV_{AC}$).

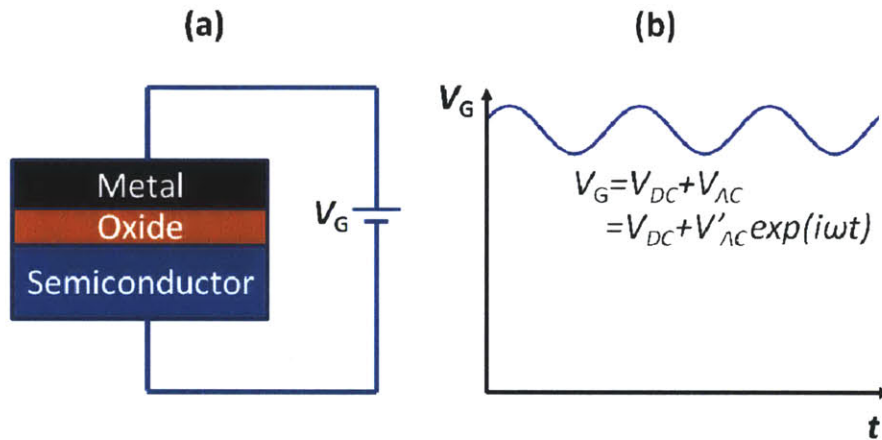


Figure 2.5 (a) MOS capacitor structure, and (b) applied voltage V_G for C-V measurement.

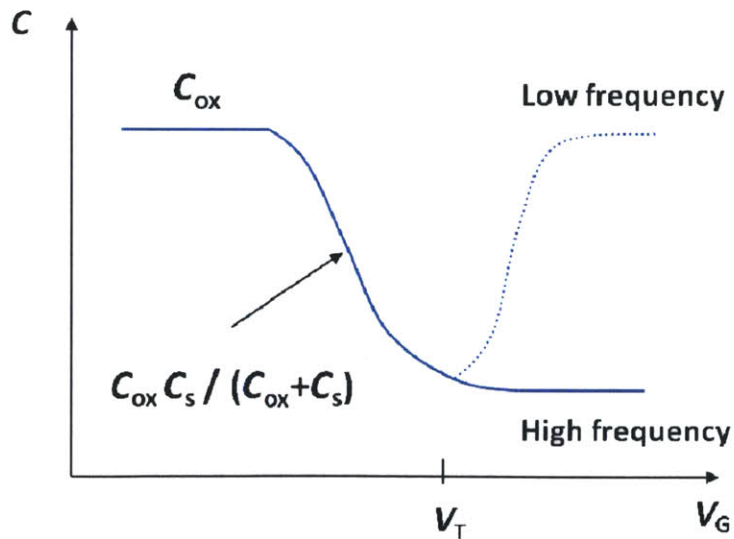


Figure 2.6 C-V characteristics of an ideal MOS capacitor on p-type substrate.

Figure 2.6 gives the C-V characteristics of an ideal MOS capacitor on a p-type substrate. The total capacitance (C) of a MOSCAP is the series capacitance of the oxide capacitance (C_{ox}) and semiconductor capacitance (C_s), which can be expressed as $C = C_{ox}C_s/(C_{ox} + C_s)$. For a negative DC bias, the holes induced by the negative DC bias accumulate near the oxide/semiconductor interface and this is called accumulation. Since the accumulation layer is very thin, the total capacitance is close to C_{ox} . When the DC bias increases, the holes start to get depleted and form a depletion region. The total capacitance becomes the series capacitance of the oxide capacitance and the depletion layer capacitance, where the depletion layer capacitance is determined by $C_s = \epsilon\epsilon_0 A/W_d$ with ϵ as the dielectric constant of the semiconductor and W_d as the width of the depletion layer. The DC bias keeps increasing until the Fermi level at the interface is closer to the conduction band and the number of electrons starts to exceed that of the holes, where the MOS capacitor enters the inversion regime. The voltage at which the MOS capacitor gets into inversion is called the threshold voltage (V_T). The inversion charges (electrons here for p-type) can shield the DC bias and prevent the depletion region from expanding. Therefore, further increase of the DC bias will only increase the inversion charge density but not the width of the depletion region. In the inversion regime, if the frequency of the AC bias is low, the inversion layer charge can keep up with the AC bias and the total capacity becomes the series capacitance of the oxide capacitance and the inversion layer capacitance. Since the inversion layer is very thin, the total capacitance is very close to the oxide capacitance. However, if the frequency of the AC bias is high, the inversion layer charge cannot keep up with the AC bias and the

total capacitance becomes the series capacitance of the oxide capacitance and the depletion layer capacitance. Since the width of the depletion layer is typically large, which makes the depletion layer capacitance fairly small, the total capacitance becomes much smaller than the oxide capacitance C_{ox} .

In a real MOS capacitor structure, because of the imperfection of the oxide-semiconductor interface, there are charged interfacial traps (Figure 2.7 (a)). C. W. Cheng theoretically simulated the effect of interfacial defects on the C-V characteristics of a MOS capacitor [2.7]. In general, there are two effects, one is the “stretch-out” of the C-V curves as shown in Figure 2.7 (b), and the other one is the frequency dispersion of C-V curves as shown in Figure 2.7 (c) [2.7].

For a real MOS capacitor, as the DC bias sweeps from accumulation to inversion, the electrons need to fill the interfacial defect states first. Therefore higher voltage is needed to get to the same level depletion or inversion and this gives the “stretch out” of C-V curve (Figure 2.7 (b)). At high AC frequency, charging and discharging of the interfacial defect states does not respond to the AC voltage, while at low AC frequency, charging and discharging the interfacial defect states partially or fully responds to the AC voltage, which adds a parallel capacitance to the semiconductor capacitance and increases the total capacitance. This causes a shift of the C-V curve for different frequencies (Figure 2.7 (c)) and is called frequency dispersion. In general, a higher defect density will give a larger stretch-out and frequency dispersion. This can be used to empirically estimate the interfacial defect density.

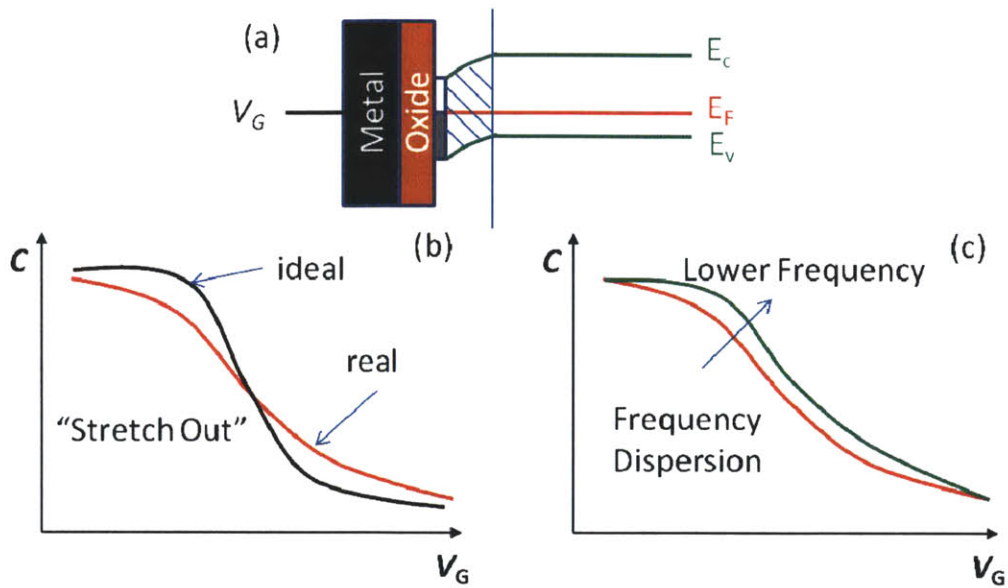


Figure 2.7 (a) Band alignment, (b) C-V characteristics showing “stretch out”, and (c) C-V characteristics showing frequency dispersion of a realistic MOS capacitor on a p-type substrate.

The interfacial defect density can be quantitatively determined by the conductance-frequency method [2.14]. The basic idea of the conductance-frequency method is that defect states at different energy levels have different response frequencies and once the frequency of the AC bias matches their response frequency, the measured conductance will give a maximum and this allows us to get the defect density at that energy level. For details in extracting the defect density by a conductance-frequency method, please read the above reference [2.14] or Chengwei’s thesis [2.7].

2.3.2 Hall Measurements

Resistivity and Hall effect [2.15] measurements are very powerful techniques to examine the transport properties (such as resistivity, carrier density, and mobility) of various materials. Typically, an I-V measurement in the absence of a magnetic field (resistivity measurement) gives the resistance of the sample and an I-V measurement in the presence of a magnetic field (Hall measurement) gives the carrier density of the sample. From the resistance and carrier density, one can easily get the carrier mobility. In this thesis, the resistivity measurement and Hall measurement together are called Hall measurement.

In the semiconductor industry, the mobility can be inferred from transistor characteristics; however a Hall measurement gives a more convenient and direct measurement of the mobility. Two sample configurations are commonly used for Hall measurement, Hall Bar and Van der Pauw, which will be discussed in detail below. One can also apply a gate voltage on the Hall structure (either on Hall bar or Van der Pauw) to control the carrier density in the material and then do the Hall measurement; this is called a gated Hall measurement.

In this thesis, the mobility for ungated structures was mainly obtained from a Van der Pauw measurement; the gated Hall measurement was mainly based on the Hall bar geometry. For the same structure, a Van der Pauw measurement and a Hall bar measurement gave very close results.

2.3.2.1 Hall Bar Configuration

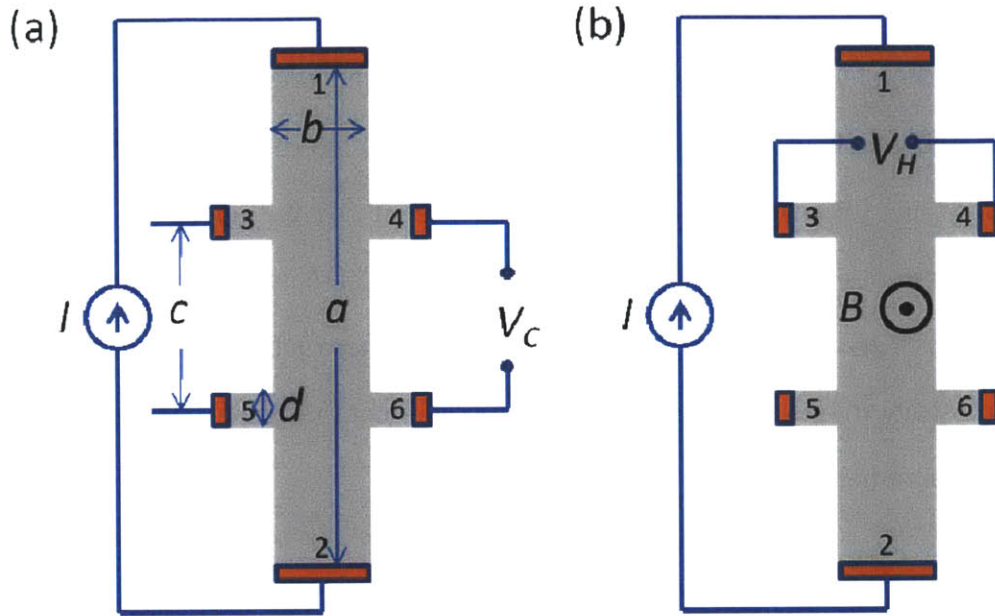


Figure 2.8 Measuring (a) resistance, and (b) carrier density using Hall bar geometry.

Figure 2.8 shows the commonly used Hall bar structure. The resistance of the sample can be obtained by applying a current between contacts 1 and 2 and measuring the voltage between contacts 4 and 6, as shown in Figure 2.8 (a).

The sheet resistance R_s is then given by:

$$R_s = \frac{V_{C46}}{I_{12}} * \frac{b}{c}, (\text{equation 2.1})$$

where R_s is the sheet resistance, V_{C46} is the measured voltage between contacts 4 and 6, I_{12} is the applied current between contacts 1 and 2, and b and c are indicated in Figure 2.8 (a). The carrier density of the sample can be obtained by applying a magnetic field perpendicular to the sample surface and a current between contacts 1 and 2, and measuring the voltage between 3 and 4, as shown in Figure 2.8 (b). The sheet carrier density can then be

obtained from the formula:

$$\frac{1}{p_s e} = \frac{V_{H34}}{I_{12} B} \text{ for } p - \text{type}; (\text{equation 2.2})$$

$$-\frac{1}{n_s e} = \frac{V_{H34}}{I_{12} B} \text{ for } n - \text{type}; (\text{equation 2.3})$$

where p_s is the sheet hole density, n_s is the sheet electron density, V_{H34} is the measured voltage between contact 3 and 4, I_{12} is the applied current between contacts 1 and 2, and B is the magnetic field.

From the sheet resistance and the sheet carrier density, one can easily get the carrier mobility from the following formula:

$$\mu_H = \frac{1}{p_s e R_s} \text{ for } p - \text{type}; (\text{equation 2.4})$$

$$\mu_H = \frac{1}{n_s e R_s} \text{ for } n - \text{type}; (\text{equation 2.5})$$

where μ_H is the hole mobility, p_s is the sheet hole density, n_s is the sheet electron density, and R_s is the sheet resistance.

To increase the accuracy of the Hall bar measurement, one needs to increase the ratio of the sample length a to sample width b , and increase the ratio of the sample width b to contact width d . In this thesis, the sample length a is 2000 μm , the sample width b is 100 μm , and the contact width d is 30 μm . This gives an error of 12.7% [2.16].

To control the carrier density in the material, a gate metal can be deposited on the Hall bar structure and a voltage can be applied between the gate metal and contact 2 as shown in Figure 2.9. The Hall measurement can then be

done in the same way as that for the ungated structure.

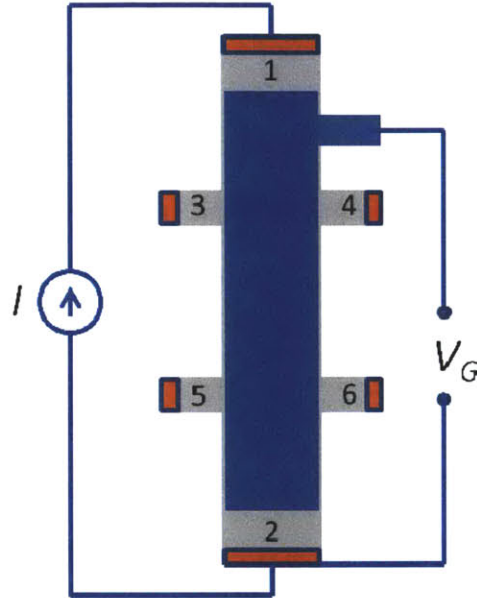


Figure 2.9 Gated Hall measurement in the Hall bar geometry.

2.3.2.2 Van der Paul Configuration

In 1958, Van der Pauw gave a method of measuring resistivity and Hall effect of discs of arbitrary shape [2.17]. Figure 2.10 shows one of the commonly used Van der Pauw structures (square shape) for Hall measurement. Since the above structure is symmetric, one can easily get the resistance by applying a current between contacts 1 and 2 and measuring the voltage between contacts 3 and 4, as shown in Figure 2.10 (a). The sheet resistance is then given by:

$$R_s = \frac{\pi}{\ln(2)} * \frac{V_{C43}}{I_{12}}. \text{ (Equation 2.6)}$$

The carrier density of the sample can be obtained by applying a magnetic field

perpendicular to the sample surface and a current between contacts 1 and 3 and measuring the voltage between 2 and 4, as shown in Figure 2.10 (b). The sheet carrier density can then be obtained from the formula:

$$\frac{1}{p_s e} = \frac{V_{H24}}{I_{13} B} \text{ for } p - \text{type}; (\text{Equation 2.7})$$

$$-\frac{1}{n_s e} = \frac{V_{H24}}{I_{13} B} \text{ for } n - \text{type}. (\text{Equation 2.8})$$

Carrier mobility can be easily calculated from equation 2.4 and 2.5.

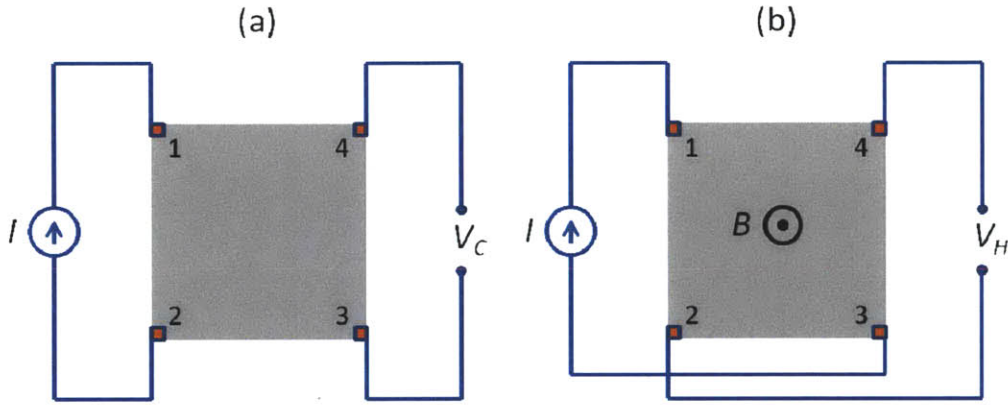


Figure 2.10 Measuring (a) resistivity, and (b) Hall coefficient using van der Pauw geometry.

There are two methods that can be used to increase the accuracy of the Van der Pauw Hall measurement. One is to reduce the contact size. For square samples with a side length of a and square contacts with a side length of b , the measurement error is less than 10% if $b/a < 0.1$ [2.18]. In this thesis, we kept $b/a = 0.075$, which gives an error around 7.5%. The other way to increase the accuracy is to do the measurement with different sets of contacts and then take the average of the measured data. This can cancel out slight

inhomogeneity in the sample.

Chapter 3 Lattice Engineered Substrate

3.1 Introduction

As we have discussed in Chapter 1, InP and semiconductor alloys lattice-matched (or slightly lattice-mismatched) to InP enable various state-of-the-art electronic and optical devices, like high speed HBTs [3.1, 3.2], HEMTs [3.3], MOSFETs [3.4], quantum-well MOSFETs [3.5] as well as 1.3 and 1.55 μm light sources and detectors.

Our research is motivated by the desire to integrate InP-based devices on Si substrates, specifically by engineering the lattice constant from Si to that of InP. There are several technology challenges in transitioning from the Si lattice constant to that of InP that can be broken down into three major challenges. The major challenges encompass (1) bridging the lattice constant of Si to that of Ge via a SiGe graded layer and wafer bonding technology, (2) the accommodation of the group IV/III-V compound nonpolar/polar interface, and (3) the engineering of the lattice constant from GaAs to that of InP. Taking a 6° GOI wafer as the growth template, we successfully demonstrated GaAs on Si with a very low threading dislocation density, no anti-phase disorder, and thin epi-film as shown in Section 1.2.2.3, Figure 1.8. Therefore, here we focus primarily on the latter challenge, the engineering of the lattice constant from that of GaAs to InP.

Specifically, our objective is to establish the InP lattice constant on (001) GaAs substrates with a 6° miscut, while maintaining a low threading dislocation density ($\sim 10^6 \text{ cm}^{-2}$) and a surface morphology suitable for eventual

device processing. The specification of the 6° miscut is important because the suppression and elimination of antiphase disorder during the growth of III-V compounds on group IV semiconductors requires the step structure provided by the miscut [3.6]. Therefore, eventual integration of the InP lattice constant with Si substrates requires an understanding of how to integrate the InP lattice on 6° offcut GaAs.

3.2. Background

Previously, Liao *et al.* have shown InP layers grown directly on GaAs (001) with 10° offcut toward <111> direction with a measured threading dislocation density of $5 \times 10^7 \text{ cm}^{-2}$ by MOCVD [3.7]. Morales *et al.* also showed the direct growth of InP on on-axis GaAs (001) via molecular beam epitaxy (MBE) with a resulting threading dislocation density of 10^8 cm^{-2} [3.8]. These results are not surprising as the direct growth of highly lattice-mismatched systems InP on GaAs, or GaAs on Si leads to large strains (>3.5%) at the mismatched interface, which result in the nucleation of large quantities of dislocations. It is well known that dislocations act as the scattering centers for free carriers, reducing the carrier mobility, and they reduce the minority carrier lifetime by introducing trap states in the middle of the energy gap, which affects the performance of minority carrier devices. Therefore, it is essential to reduce the threading dislocation density if InP-based devices are to be considered for integration with GaAs or Si substrates. Compositionally graded buffers have been shown to reduce the threading dislocation density for InP layers grown on GaAs. Quitoriano and Fitzgerald established InP on on-axis GaAs (001)

substrates with threading dislocation density of $1.2 \times 10^6 \text{ cm}^{-2}$ by using a combination of graded $\text{In}_x\text{Ga}_{1-x}\text{As}$ and $\text{In}_y\text{Ga}_{1-y}\text{P}$, grown via atmospheric pressure MOCVD [3.9]. However, they found that the identical compositionally graded layers grown on offcut GaAs substrates were more prone to phase separation (undulation of material composition) [3.9], which promoted surface roughening and a higher threading dislocation density. Although less explored, another path to get to the InP lattice constant would be to use compositionally graded $\text{GaAs}_z\text{Sb}_{1-z}$ alloys. Chen *et al.* demonstrated $\text{GaAs}_z\text{Sb}_{1-z}$ compositionally step-graded buffers with up to 58% Sb concentration on on-axis GaAs (001) with threading dislocation density of $1 \times 10^9 \text{ cm}^{-2}$ [3.10].

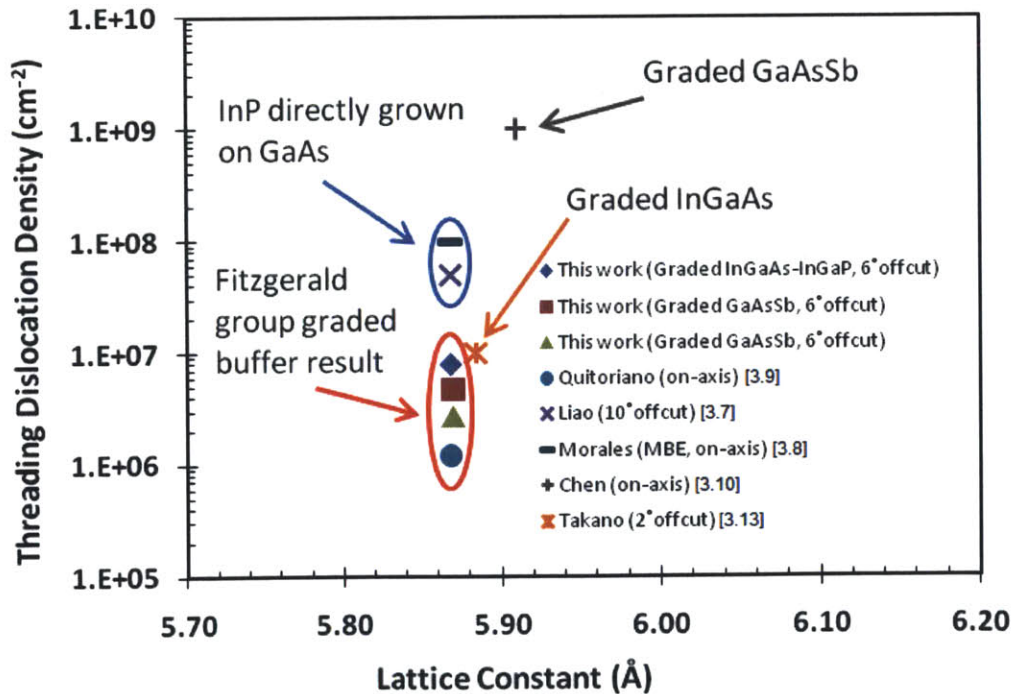


Figure 3.1 Summary of threading dislocation density results for films with up to InP lattice constant on a GaAs substrate in the literature and this work [3.7-3.10, 3.13].

Table 3.1 Characteristics of the graded buffers fabricated in this study.

#	GaAs Substrate	Structure	Lattice Constant (Å)	Growth gradient (%strain/ μm)	Threading Dislocation Density (cm^{-2})	RMS (nm)
1	On-axis	(700°C) $\text{In}_x\text{Ga}_{1-x}\text{As}$ ($x=0-0.30$)	5.775	1.05	1.9×10^6	7.5
2	6° offcut	(700°C) $\text{In}_x\text{Ga}_{1-x}\text{As}$ ($x=0-0.30$)	5.775	1.05	$\sim 5 \times 10^8$	26.1
3	6° offcut	(450°C) $\text{In}_x\text{Ga}_{1-x}\text{As}$ ($x=0.10-0.30$)/ (700°C) $\text{In}_x\text{Ga}_{1-x}\text{As}$ ($x=0-0.10$)	5.775	2.09/0.35	1.4×10^6	7.5
4	6° offcut	(650°C) $\text{In}_y\text{Ga}_{1-y}\text{P}$ ($y=0.78-1.00$)/ (450°C) $\text{In}_x\text{Ga}_{1-x}\text{As}$ ($x=0.10-0.30$)/ (700°C) $\text{In}_x\text{Ga}_{1-x}\text{As}$ ($x=0-0.10$)	5.869	0.80/2.43/1.29	7.9×10^6	33.0
5	6° offcut	(575°C) $\text{GaAs}_{1-z}\text{Sb}_z$ ($z=0-0.49$)	5.870	0.64	2.7×10^6	
6	6° offcut	(575°C) $\text{GaAs}_{1-z}\text{Sb}_z$ ($z=0-0.49$)	5.870	1.06	4.7×10^6	7.4
7	6° offcut	(575°C) $\text{GaAs}_{1-z}\text{Sb}_z$ ($z=0-0.49$)	5.870	1.63	3.3×10^7	

* The structure and growth gradient in this work are labeled in the order of top/middle/bottom.

In this thesis, we explore and compare both tandem $\text{In}_x\text{Ga}_{1-x}\text{As}$ - $\text{In}_y\text{Ga}_{1-y}\text{P}$ and $\text{GaAs}_z\text{Sb}_{1-z}$ graded buffers to attain an InP lattice constant on (001) GaAs offcut 6° to the $\langle 111 \rangle \text{A}$. The objective was to establish compositionally graded buffer technology that yielded a low threading dislocation density with suitable surface morphology, while maintaining an overall epitaxial layer thickness that was less than $5.0 \mu\text{m}$ to avoid possible cracking caused by thermal expansion mismatch [3.11]. Figure 3.1 summarizes the threading dislocation density results from the previous work done for films with up to InP lattice constant on a GaAs substrate along with the results from this work. The threading dislocation density results reported in this work are slightly higher than the values reported by Quitoriano and Fitzgerald as the best results on on-axis GaAs [3.9]; however, they are the best results reported to date on 6° offcut (001) GaAs. Table 3.1 summarizes the essential characteristics of the

structures we fabricated and studied for this survey that we will discuss in detail.

3.3. Compositionally-Graded InGaAs-InGaP Alloys for Achieving InP on offcut GaAs

In this section, we introduce our experimental results on compositionally-graded InGaAs-InGaP Alloys for achieving InP on offcut GaAs. This tandem approach was found to be necessary because phase separation in the InGaAs alloys leads to surface roughening and a high threading dislocation density when grading to lattice constants greater than that of $\text{In}_{0.30}\text{Ga}_{0.70}\text{As}$. To get to the InP lattice, we first tried to optimize the $\text{In}_x\text{Ga}_{1-x}\text{As}$ graded buffer growth conditions to get good quality $\text{In}_{0.30}\text{Ga}_{0.70}\text{As}$; then we optimized the gas switching sequence to establish a good quality $\text{In}_{0.78}\text{Ga}_{0.32}\text{P}/\text{In}_{0.30}\text{Ga}_{0.70}\text{As}$ interface; finally we implemented a graded $\text{In}_y\text{Ga}_{1-y}\text{P}$ system to continue grading the lattice constant from $\text{In}_{0.30}\text{Ga}_{0.70}\text{As}$ to InP.

3.3.1 Experimental Methods

The compositionally-graded InGaAs-InGaP alloys presented in this thesis were grown on on-axis or 6° offcut toward $\langle 111 \rangle$ A GaAs (001) substrates purchased from AXT. The epitaxy was conducted using a Thomas Swan/AIXTRON low pressure MOCVD reactor with a close-coupled showerhead. Trimethylgallium (TMGa), trimethylindium (TMIn), arsine (AsH_3) and phosphine (PH_3) were used as the precursors and no dopants were used.

The growth pressure was 100 Torr and purified N₂ was used as the carrier gas. Growth temperatures ranged from 450°C-700°C, which was the actual wafer surface temperature based upon an optical pyrometry calibration. At the beginning of each growth, the temperature was ramped to the graded buffer growth temperature with an AsH₃ over pressure, and a 300 nm GaAs homoepitaxial layer was grown before the growth of graded buffers to minimize any effects of substrate surface impurities. For a graded buffer, the growth gradient, also called the mismatch gradient, was defined as the mismatch introduced per unit film thickness with a unit of [strain/μm]. In this work, we introduced mismatch continuously with a constant growth gradient for each graded buffer section. The structure and growth gradient for the graded buffer are both labeled in the order of top/middle/bottom.

The films were characterized with cross-sectional (X) and plan-view (PV) transmission electron microscopy (TEM), x-ray diffraction (XRD), and atomic force microscopy (AFM) techniques. The TEM samples were prepared with traditional mechanical polishing and argon ion milling techniques and the samples were imaged with either a JEOL 200CX or a JEOL 2011 microscope operated at 200 kV. The threading dislocation density was determined by PVTEM. The single PVTEM image shown in the figures does not give the overall threading dislocation density of that sample. For material with a threading dislocation density $<1 \times 10^7 \text{ cm}^{-2}$, the density was determined after viewing 30 PVTEM images (670 μm² of area) in a $g = \langle 0 \ 2 \ 2 \rangle$ diffraction condition. This gives the threading dislocation density an uncertainty range (95% confidence interval) of less than 1/3 of the density. For material with a

threading dislocation density $> 1 \times 10^8 \text{ cm}^{-2}$, the density was estimated from 2 PVTEM images. PANalytical X'pert Pro and Bruker D8 XRD systems were used to measure the lattice constant and to determine the alloy composition. The surface morphology and roughness were measured with a Veeco Nanoscope IV AFM operating in the tapping mode.

3.3.2 InGaAs Graded Buffer Grown at High Temperature

Quitoriano and Fitzgerald demonstrated InP on on-axis (001) GaAs with a threading dislocation density $= 1.2 \times 10^6 \text{ cm}^{-2}$ by growing hybrid $\text{In}_x\text{Ga}_{1-x}\text{As}$ and $\text{In}_y\text{Ga}_{1-y}\text{P}$ graded buffers [3.9]. Lee and Fitzgerald demonstrated up to $X_{\text{In}}=0.35$ $\text{In}_x\text{Ga}_{1-x}\text{As}$ graded buffers on on-axis (001) GaAs substrates with a threading dislocation density $< 9.5 \times 10^4 \text{ cm}^{-2}$ with a growth gradient of 0.51% strain/ μm [3.12]. In both of their experiments, the $\text{In}_x\text{Ga}_{1-x}\text{As}$ was grown on on-axis substrates at high temperatures (700°C-725°C) to prevent the formation of phase separated regions.

We first tried to understand the results of applying previously successful compositionally graded buffer schemes on offcut GaAs substrates. As a first step, we grew $\text{In}_x\text{Ga}_{1-x}\text{As}$ graded buffers at 700 °C with targeted In concentrations of $X_{\text{In}}=0.30$ on on-axis (001) GaAs and (001) GaAs miscut 6° toward the $\langle 111 \rangle \text{A}$. Each graded buffer was topped with a uniform $\text{In}_{0.30}\text{Ga}_{0.70}\text{As}$ cap. Figure 3.2 (a) and (b) show X-TEM images of the two films grown on different GaAs substrate orientations. No phase separation was observed for the InGaAs grown on on-axis GaAs substrates (Figure 3.2 (a))

and the threading dislocation density is low as shown in the PV-TEM in Figure 3.2 (c); however, the InGaAs grown on 6° offcut GaAs shows very serious phase separation (Figure 3.2 (b)) and a much higher threading dislocation density (Figure 3.2 (d)). We determined that phase separation in InGaAs graded buffers worsens with substrate offcut. This is consistent with a result demonstrated by Quitoriano and Fitzgerald, which showed that phase separation of InGaAs occurs more easily if grown on 2° offcut toward <111> GaAs substrate compared to grown on on-axis GaAs [3.9].

Phase separation, which is the compositional fluctuations/undulations at the microscopic level, can lead to significant local strain variations. The strain variations have two deleterious effects. First, the strain can pin (stop) dislocation glide and lead to further dislocation nucleation, resulting in a high threading dislocation density [3.9]. Second, it will affect local growth rates and film incorporation properties for different regions of the surface and consequently increase the surface roughness. Figure 3.2 (e) and (f) show AFM topographic images and the corresponding root-mean-square (RMS) surface roughness for the InGaAs graded buffers grown on the different orientation GaAs substrates. InGaAs grown on 6° offcut toward <111>A GaAs shows a much higher roughness (Figure 3.2 (f)) compared to InGaAs grown on on-axis GaAs (Figure 3.2 (e)). InGaAs grown on 6° offcut toward <111>A GaAs also shows branch-like patterns with peak-to-peak period of 1.3 μm (Figure 3.2 (f)), which confirms the existence of phase separation in this film. These results confirm our TEM findings that the InGaAs material quality is heavily dependent on substrate orientation.

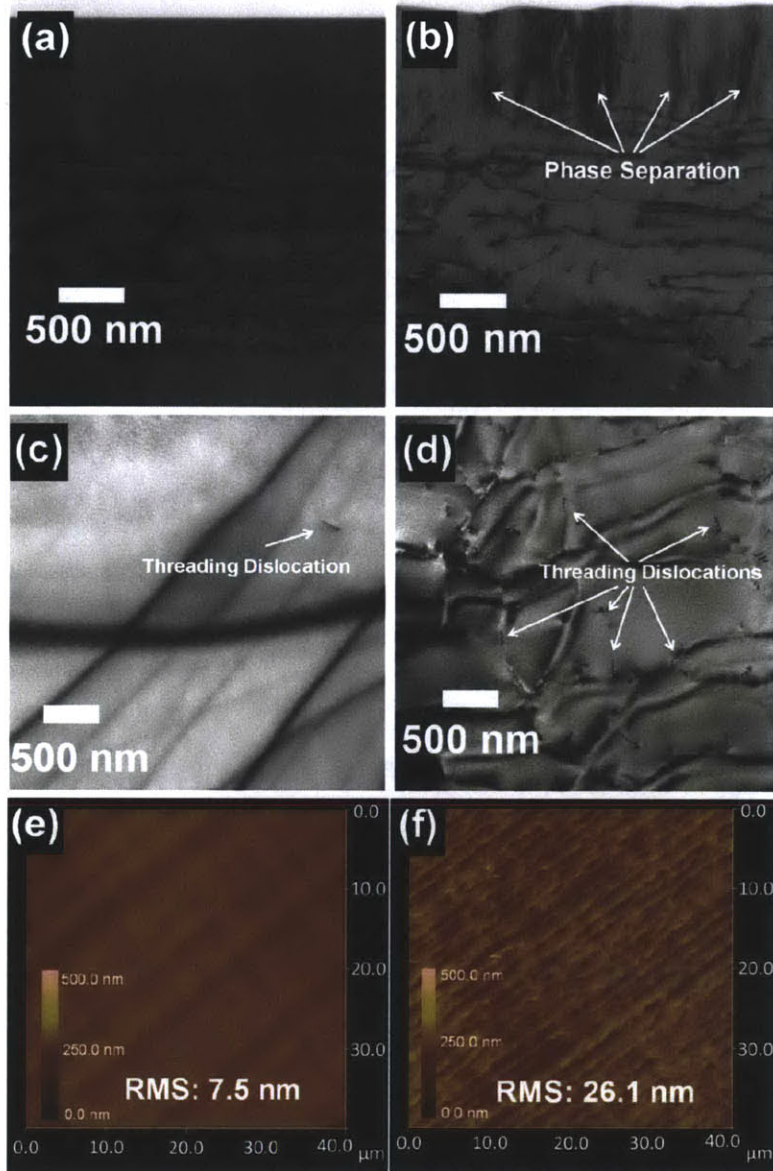


Figure 3.2 X-TEM images ($g=\langle 0\ 2\ 2 \rangle$) of (700°C) $\text{In}_x\text{Ga}_{1-x}\text{As}$ ($x=0-0.30$) graded buffers grown on (a) on-axis GaAs (001) substrate and (b) 6° offcut toward $\langle 111 \rangle$ A GaAs (001) substrates. PV-TEM images ($g=\langle 0\ 2\ 2 \rangle$) of (700°C) $\text{In}_x\text{Ga}_{1-x}\text{As}$ ($x=0-0.30$) graded buffers grown on (c) on-axis GaAs (001) substrate and (d) 6° offcut toward $\langle 111 \rangle$ A GaAs (001) substrates. AFM images of (700°C) $\text{In}_x\text{Ga}_{1-x}\text{As}$ ($x=0-0.30$) graded buffers grown on (e) on-axis GaAs (001) substrate and (f) 6° offcut toward $\langle 111 \rangle$ A GaAs (001) substrates. Phase separation exists for the sample grown on 6° offcut substrate.

3.3.3 InGaAs Graded Buffer Grown at Low Temperature

Phase separation phenomena occur during the growth of InGaAs graded buffers due to the surface kinetics during epitaxy and the non-equilibrium nature of MOCVD. To decrease the surface kinetics of phase separation, one approach is to lower the growth temperature to decrease the surface diffusivity. Takano *et al.* [3.13] grew $\text{In}_x\text{Ga}_{1-x}\text{As}$ graded buffers on GaAs (001) misoriented 2° toward $\langle 111 \rangle_A$ to $X_{\text{In}}=0.57$ without any phase separation at 450°C and the threading dislocation density was measured as $1 \times 10^7 \text{ cm}^{-2}$.

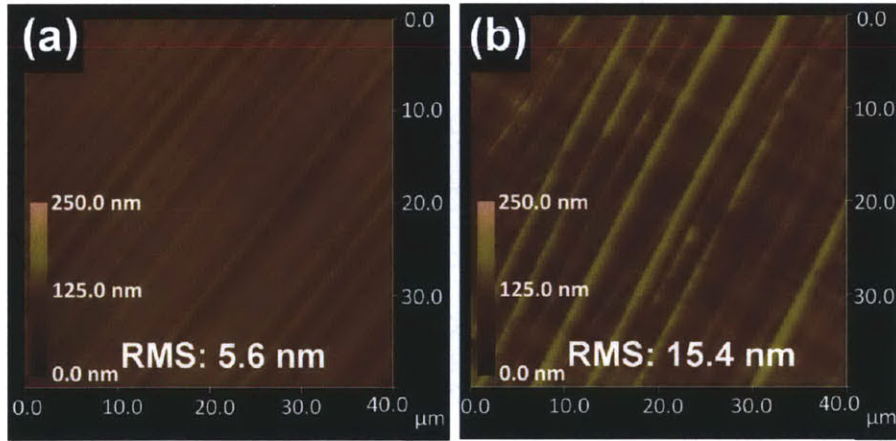


Figure 3.3 AFM images of (450°C) $\text{In}_x\text{Ga}_{1-x}\text{As}$ ($x=0-0.33$) graded buffers on (a) on-axis GaAs (001) substrate and (b) 6° offcut toward $\langle 111 \rangle_A$ GaAs (001) substrate.

We also tried to apply this low temperature graded buffer scheme on 6° offcut GaAs substrates. We grew $\text{In}_x\text{Ga}_{1-x}\text{As}$ graded buffers at 450°C with a final In concentration of $X_{\text{In}}=0.33$ on on-axis (001) GaAs and (001) GaAs miscut 6°

toward $\langle 111 \rangle_A$. Figure 3.3 (a) and (b) show AFM images of the two films grown on GaAs substrates with different orientations. InGaAs grown on 6° offcut GaAs substrates (Figure 3.3 (b)) shows a much rougher surface than InGaAs grown on on-axis GaAs substrates (Figure 3.3 (a)). The rough surface is likely due to the reduced dislocation glide for low temperature growth, which led to the heavier surface crosshatched regions that are indicative of dislocation pile-up formation. Therefore, we concluded that low temperature compositional grading is not effective for 6° offcut GaAs substrates.

3.3.4 InGaAs Graded Buffer Grown at Low/High Temperatures

Since the onset of phase separation occurs with higher In content films, we separated the graded layer profile with the following structure: $(650^\circ\text{C})\text{In}_y\text{Ga}_{1-y}\text{P}(y=0.78-1.00)/(450^\circ\text{C})\text{In}_x\text{Ga}_{1-x}\text{As}(x=0.10-0.30)/(700^\circ\text{C})\text{In}_x\text{Ga}_{1-x}\text{As}(x=0-0.10)/\text{GaAs}$. This structure suppresses phase separation, while suppressing dislocation pile-up formation, by maintaining a high growth temperature for low indium content $\text{In}_x\text{Ga}_{1-x}\text{As}$, then reducing the growth temperature for the higher In content $\text{In}_x\text{Ga}_{1-x}\text{As}$, and finally transitioning to the compositional grading to the $\text{In}_y\text{Ga}_{1-y}\text{P}$ a materials system, which is more immune to phase separation at a higher growth temperature, which also facilitates dislocation glide. InGaP is more immune to phase separation at the lattice-matched transition composition since the InGaP alloy will be In-rich and far from the 50-50 In-Ga composition where surface-driven phase separation is most problematic.

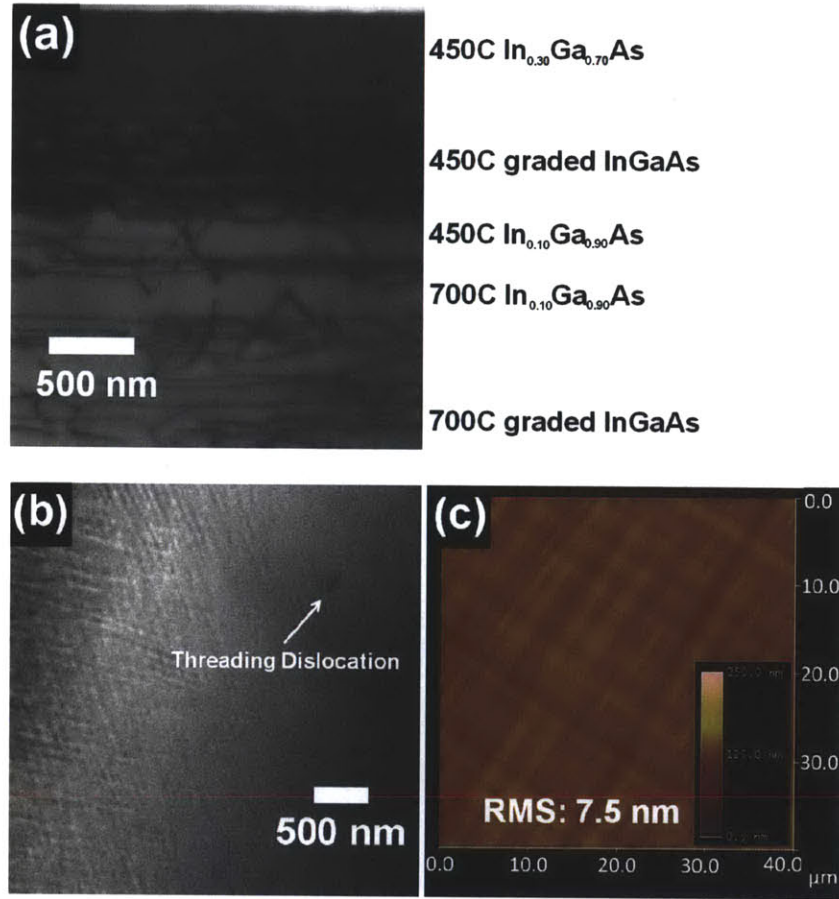


Figure 3.4 (a) X-TEM ($g=\langle 0\ 2\ 2 \rangle$), (b) PV-TEM ($g=\langle 0\ 2\ 2 \rangle$), and (c) AFM of $(450^\circ\text{C})\text{In}_x\text{Ga}_{1-x}\text{As}$ ($x=0.10-0.30$)/ $(700^\circ\text{C})\text{In}_x\text{Ga}_{1-x}\text{As}$ ($x=0-0.10$) graded buffer on 6° offcut toward $\langle 111 \rangle$ A GaAs (001) substrate.

We examined the grading profile mentioned in the previous paragraph in part by first examining the combination of the high temperature and low temperature $\text{In}_x\text{Ga}_{1-x}\text{As}$ compositional grading. Figure 3.4 shows the X-TEM, PV-TEM and the AFM images of $(450^\circ\text{C})\text{In}_y\text{Ga}_{1-y}\text{As}$ ($y=0.10-0.30$)/ $(700^\circ\text{C})\text{In}_x\text{Ga}_{1-x}\text{As}$ ($x=0-0.10$). From the PVTEM, one still can see the contrast caused by phase separation. However, the phase separation in this sample was not so serious that it significantly blocked dislocation glide. This

structure exhibited a much lower threading dislocation density ($1.4 \times 10^6 \text{ cm}^{-2}$) compared to the threading dislocation density ($\sim 5 \times 10^8 \text{ cm}^{-2}$) measured for the $\text{In}_x\text{Ga}_{1-x}\text{As}$ ($x=0-0.30$) grown exclusively at 700°C . The surface roughness was also lower for this sample compared to that of $\text{In}_x\text{Ga}_{1-x}\text{As}$ ($x=0-0.30$) grown at 700°C (Figure 3.2 (f)) or $\text{In}_x\text{Ga}_{1-x}\text{As}$ ($x=0-0.33$) grown at 450°C (Figure 3.3 (b)). In general, we found a suitable solution for grading $\text{In}_x\text{Ga}_{1-x}\text{As}$ on offcut GaAs up to $X_{\text{In}}=0.30$ and conducted experiments to extend the lattice constant to that of InP by implementing $\text{In}_y\text{Ga}_{1-y}\text{P}$ compositional grading past this point.

3.3.5 InGaP/InGaAs Interface

Growth of $\text{In}_y\text{Ga}_{1-y}\text{P}$ at 650°C immediately after terminating the growth of $\text{In}_x\text{Ga}_{1-x}\text{As}$ at 450°C is non-trivial. Two gas switching methods were investigated for the initiation of InGaP on InGaAs. For the first method, after the growth of InGaAs at 450°C was complete, the temperature was increased from 450°C to 650°C under an AsH_3 overpressure, and then lattice-matched InGaP was grown at 650°C on top of InGaAs. This procedure resulted in a very defective InGaP/InGaAs interface, as shown in Figure 3.5 (a). The other approach introduced a 40 nm lattice-matched InGaP layer at 450°C immediately after the growth of InGaAs at 450°C , and the temperature was then increased to 650°C under a PH_3 overpressure, where the InGaP growth was continued. The InGaP/InGaAs interface was dramatically improved for the second approach interface, as shown in Figure 3.5 (b). Therefore, the latter gas switching procedure was used for the rest of our studies.

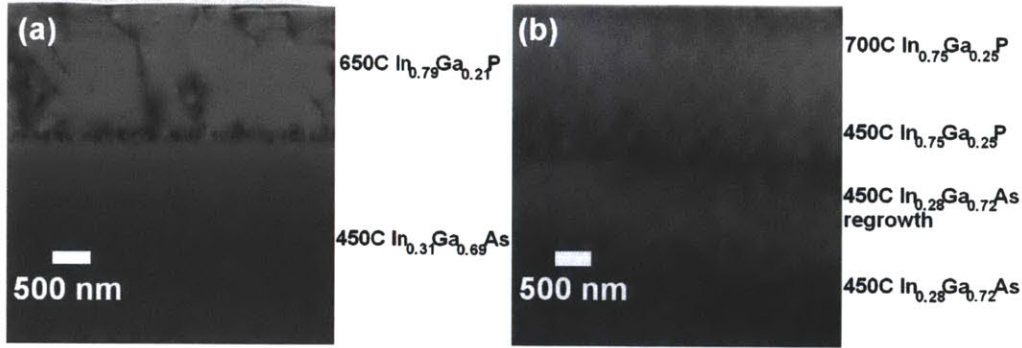


Figure 3.5 X-TEM ($g=\langle 0\ 2\ 2 \rangle$) of (a) $(650^\circ\text{C})\text{In}_{0.79}\text{Ga}_{0.21}\text{P}/(450^\circ\text{C})\text{In}_{0.31}\text{Ga}_{0.69}\text{As}$, and (b) $(650^\circ\text{C})\text{In}_{0.75}\text{Ga}_{0.25}\text{P}/(450^\circ\text{C})\text{In}_{0.75}\text{Ga}_{0.25}\text{P}/(450^\circ\text{C})\text{In}_{0.28}\text{Ga}_{0.72}\text{As}$ interfaces.

3.3.6 InGaAs-InGaP Graded Buffer up to the InP Lattice

With the complete high temperature InGaP/low temperature InGaAs/high temperature InGaAs sequence, we demonstrated highly quality InGaP films up to the InP lattice on offcut GaAs. Figure 3.6 shows the X-TEM, PV-TEM and AFM images of the final structure that we grew to fabricate the InP lattice constant on a 6° offcut toward $\langle 111 \rangle$ A GaAs substrate. First, $\text{In}_x\text{Ga}_{1-x}\text{As}$ ($x=0-0.10$) graded buffer was grown at 700°C . Then the growth temperature was decreased to 450°C under an AsH_3 over pressure. After the temperature was stabilized at 450°C , an $\text{In}_x\text{Ga}_{1-x}\text{As}$ ($x=0.10-0.30$) graded buffer was grown. After the growth of an $\text{In}_{0.30}\text{Ga}_{0.70}\text{As}$ cap layer, a 40 nm $\text{In}_{0.78}\text{Ga}_{0.22}\text{P}$ layer lattice-matched to $\text{In}_{0.30}\text{Ga}_{0.70}\text{As}$ was grown at 450°C . Then the temperature was increased to 650°C under PH_3 overpressure, and another 200 nm $\text{In}_{0.78}\text{Ga}_{0.22}\text{P}$ was grown at 650°C . Finally, $\text{In}_y\text{Ga}_{1-y}\text{P}$ was graded to the lattice constant of InP and a $1.3\ \mu\text{m}$ InP cap was grown.

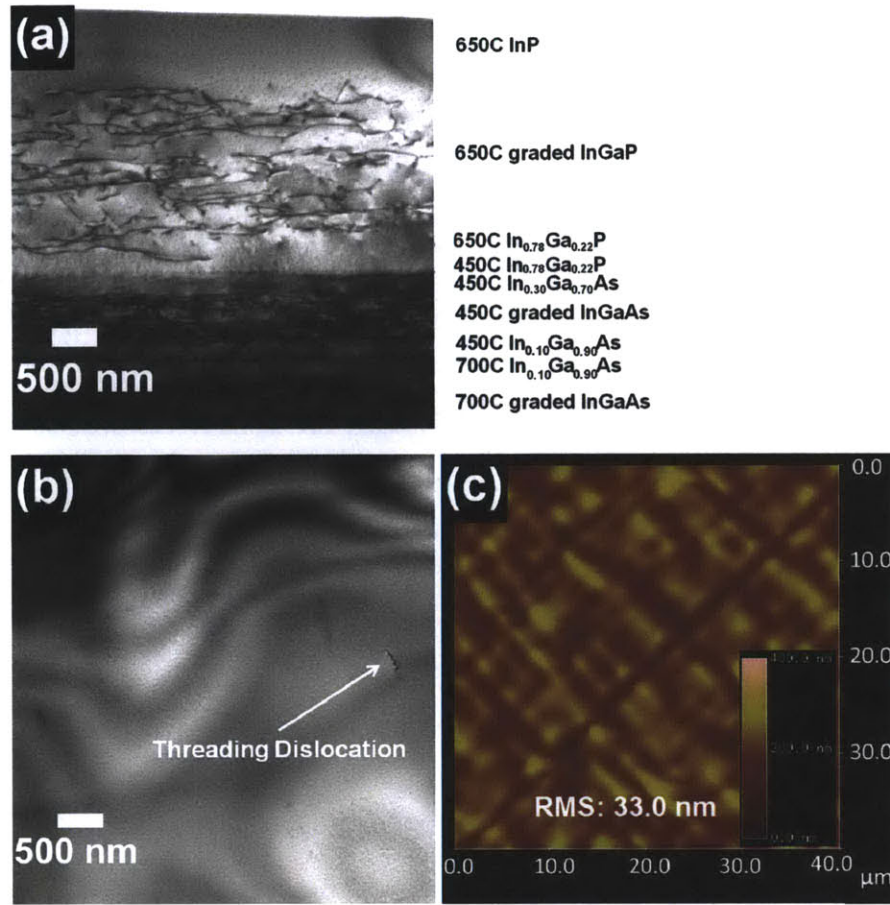


Figure 3.6 (a) X-TEM ($g=\langle 0\ 2\ 2 \rangle$), (b) PV-TEM ($g=\langle 0\ 2\ 2 \rangle$), and (c) AFM of $(650^\circ\text{C})\text{In}_x\text{Ga}_{1-x}\text{P}$ ($x=0.78-1$)/ $(450^\circ\text{C})\text{In}_{0.78}\text{Ga}_{0.22}\text{P}$ / $(450^\circ\text{C})\text{In}_x\text{Ga}_{1-x}\text{As}$ ($x=0.10-0.30$)/ $(700^\circ\text{C})\text{In}_x\text{Ga}_{1-x}\text{As}$ ($x=0-0.10$) graded buffer on a GaAs (001) substrate 6° offcut toward $\langle 111 \rangle\text{A}$.

The threading dislocation density for this sample is $7.9 \times 10^6 \text{ cm}^{-2}$ as measured from PV-TEM images (Figure 3.6 (b)). As shown in Figure 3.6 (c), the final surface roughness is 33.0 nm on a $40 \mu\text{m} \times 40 \mu\text{m}$ scale. Compared to $(700^\circ\text{C})\text{In}_x\text{Ga}_{1-x}\text{As}$ ($x=0-0.30$) grown on offcut GaAs, this sample shows crosshatch patterns with a larger peak-to-peak period ($3.6 \mu\text{m}$) in AFM, which means there is no phase separation in this film (also confirmed by TEM). That is why this sample has a lower threading dislocation density. Compared to

sample #3, the InGaAs growth gradient of this sample is higher than that of sample #3. This is because we want the total buffer thickness to be within 5 μm to avoid possible cracking if in the future we need to integrate this structure on Si. If the InGaAs growth gradient was lower, the total threading dislocation density for this sample could be lower.

3.4 Compositionally-Graded GaAsSb Alloys for Achieving InP on offcut GaAs

3.4.1 Experimental Methods

Compositionally-graded GaAsSb alloys presented in this thesis were grown on 6° offcut toward $\langle 111 \rangle_A$ GaAs (001) substrates purchased from AXT. The epitaxy was conducted using a Thomas Swan/AIXTRON low pressure MOCVD reactor with a close-coupled showerhead. Trimethylgallium (TMGa), trimethylantimony (TMSb), and arsine (AsH_3) were used as the precursors and no dopants were used. The growth pressure was 100 Torr and Pd-diffused H_2 was used as the carrier gas.

For the $\text{GaAs}_{1-z}\text{Sb}_z$ graded buffer, a growth temperature of 575°C was used based on ref [3.14]. GaAsSb growth at 650°C was also attempted but the film morphology was of lower quality. One difficulty with the $\text{GaAs}_{1-z}\text{Sb}_z$ graded buffer growth is the control of the V/III ratio. If the V/III ratio is too high, it could form Sb droplets and also greatly reduce the Sb incorporation. If the V/III ratio is too low, it could form Ga droplets. In our growth, V/III ratios of 0.8, 1.0, and

1.2 were compared and we determined that a V/III of 1.0 provided the most consistent film morphology and we used this ratio for our experimentation.

Similar to the InGaAs-InGaP growth, at the beginning of each GaAsSb growth, the temperature was ramped up to the graded buffer growth temperature with an AsH₃ over pressure and a 300 nm GaAs homoepitaxial layer was grown before the growth of graded buffers to minimize any effects of the substrate surface impurities. For a graded buffer, the growth gradient, also called the mismatch gradient, was defined as the mismatch introduced per unit film thickness with a unit of [strain/ μm]. For GaAs_{1-z}Sb_z graded buffer, we introduced mismatch continuously with a constant growth gradient.

The films were characterized with cross-sectional (X) and plan-view (PV) transmission electron microscopy (TEM), x-ray diffraction (XRD), and atomic force microscopy (AFM) techniques. The TEM samples were prepared with traditional mechanical polishing and argon ion milling techniques and the samples were imaged with either a JEOL 200CX or JEOL 2011 microscope operated at 200 kV. The threading dislocation density was determined by PVTEM measurement. The single PVTEM image showed in the figures does not give the overall threading dislocation density of that sample. The threading dislocation density was determined after viewing 30 PVTEM images (670 μm^2 of area) in a $g=\langle 0\ 2\ 2 \rangle$ diffraction condition. This gives the threading dislocation density an uncertainty range (95% confidence interval) of less than 1/3 of the density. PANalytical X'pert Pro and Bruker D8 XRD systems were used to measure the lattice constant and determine the alloy composition.

The surface morphology and roughness were measured with a Veeco Nanoscope IV AFM operating in the tapping mode.

3.4.2 GaAsSb Graded Buffer Grown at 575°C

Another system that can be compositionally graded to engineer the lattice constant of GaAs to that of InP is the $\text{GaAs}_{1-z}\text{Sb}_z$ alloy system. A $\text{GaAs}_{0.51}\text{Sb}_{0.49}$ alloy is lattice-matched to InP. Thermodynamically, $\text{GaAs}_{1-z}\text{Sb}_z$ alloys should exhibit a miscibility gap and a phase separation should be observed. However, this phase separation can be kinetically suppressed due to the small surface diffusivity of Sb. Cherng *et al.* demonstrated the growth of $\text{GaAs}_{0.50}\text{Sb}_{0.50}$, an alloy composition within the miscibility gap, on an InP substrate that did not exhibit phase separation [3.14]. Chen *et al.* demonstrated $\text{GaAs}_{1-z}\text{Sb}_z$ compositionally step-graded buffers up to $X_{\text{Sb}}=0.58$ on on-axis GaAs (001) with a threading dislocation density of $1 \times 10^9 \text{ cm}^{-2}$ [3.10]. In our study, we used continuously graded buffers instead of step-graded buffers and $\text{GaAs}_{1-z}\text{Sb}_z$ was graded to $X_{\text{Sb}}=0.49$ to acquire the InP lattice constant. The growth condition was discussed in section 3.4.1.

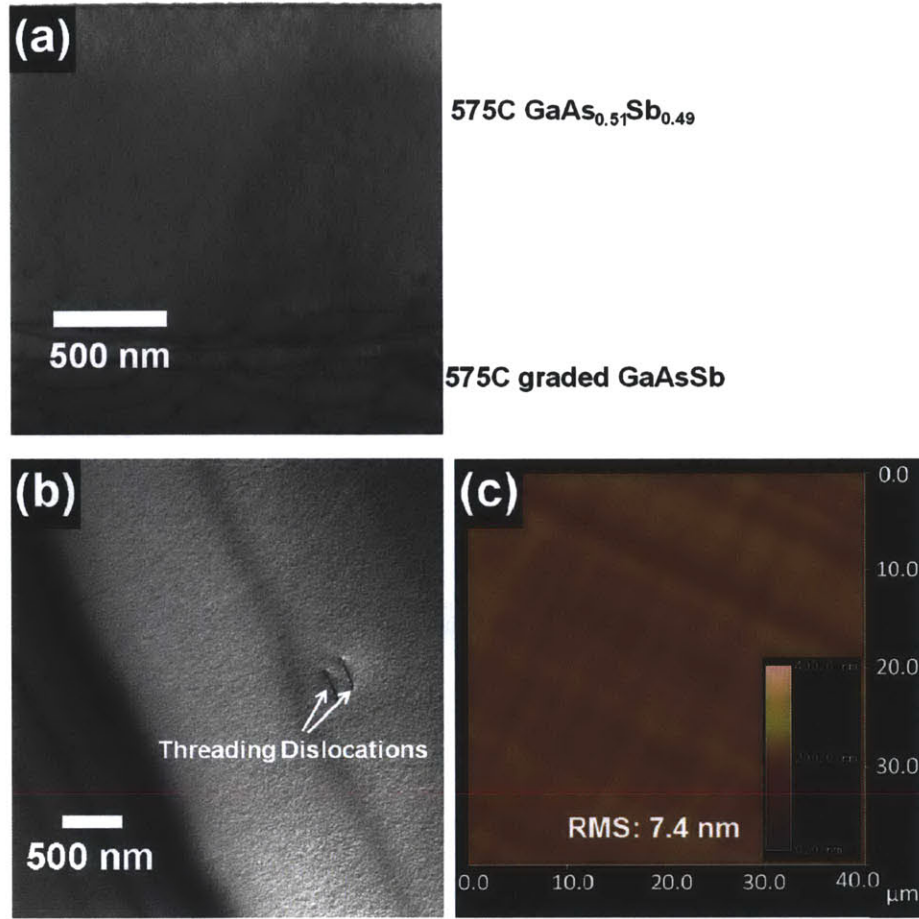


Figure 3.7 (a) X-TEM ($g=\langle 0\ 2\ 2 \rangle$), (b) PV-TEM ($g=\langle 0\ 2\ 2 \rangle$), and (c) AFM of (575°C)GaAs_{1-z}Sb_z($z=0-0.49$) graded buffer on a (001) GaAs substrate offcut 6° toward $\langle 111 \rangle_A$. No phase separation was observed in this sample.

Figure 3.7 shows the characteristics of a linearly graded GaAs_{1-z}Sb_z buffer with a cap that is lattice-matched to InP. Figure 3.7 (a) shows the XTEM image of the GaAs_{1-z}Sb_z graded buffer cap layer with a final Sb concentration of $X_{Sb}=0.49$, which is lattice matched to InP. The growth gradient was controlled to be 1.06% strain/ μm to give a buffer thickness within 5.0 μm . Figure 3.7 (b) shows the PVTEM image of this sample. The threading

dislocation density of this sample is $4.7 \times 10^6 \text{ cm}^{-2}$, which is lower than that of the tandem graded layers of $\text{In}_x\text{Ga}_{1-x}\text{As-In}_y\text{Ga}_{1-y}\text{P}$ previously presented in this thesis. Figure 3.7 (c) shows that the surface roughness of this sample is 7.4 nm on a $40 \text{ } \mu\text{m} \times 40 \text{ } \mu\text{m}$ scale, which is also much lower than that of the $\text{In}_x\text{Ga}_{1-x}\text{As/In}_y\text{Ga}_{1-y}\text{P}$ tandem graded buffers grown in this study. This low surface roughness may be due to several reasons. First, there is no phase separation in the $\text{GaAs}_{1-z}\text{Sb}_z$ graded buffer. Second, the growth temperature of the graded $\text{GaAs}_{1-z}\text{Sb}_z$ is low and typically a lower growth temperature will provide for a lower surface roughness. Thirdly, Sb can act as a surface surfactant due to its large atom radius and low vapor pressure, and Sb atoms can accumulate at the surface and alter the surface structure of many materials [3.15, 3.16]. A lower growth temperature can result in a high threading dislocation density due to the lower dislocation glide velocities, but the data suggests that the dislocation velocity is high enough to avoid a rapid rise in threading dislocation density.

3.4.3 Dislocation Kinetics for a GaAsSb Graded Buffer

We also studied the dependence of the threading dislocation density on the growth gradient (defined as the strain/thickness) for the graded $\text{GaAs}_{1-z}\text{Sb}_z$ alloy system. $\text{GaAs}_{1-z}\text{Sb}_z$ alloys were continuously graded to a 49% Sb concentration with three different growth gradients: 0.64% strain/ μm , 1.06% strain/ μm and 1.63% strain/ μm . Figure 3.8 plots the PV-TEM measured threading dislocation density values of these three samples in addition to the calculated (*i.e.*, equilibrium) threading dislocation density values based upon

dislocation propagation theory [3.17]. The following analysis only applies for continuously-graded buffers with constant growth gradients. The lattice mismatch f is always the sum of elastic strain ϵ and plastic strain δ :

$$f = \epsilon + \delta. \text{ (equation 3.1)}$$

For a large growth thickness h , the graded buffer is almost fully relaxed such that the plastic strain δ is equal to the mismatch f . Therefore, the plastic strain in a graded film of thickness h is:

$$\delta = f = C_f h, \text{ (equation 3.2)}$$

where C_f is the mismatch gradient (mismatch/unit thickness) and we call C_f the growth gradient in this thesis. On the other hand, the plastic strain allowed by kinetics, δ_k is given by:

$$\delta_k = \rho b l / 2, \text{ (equation 3.3)}$$

where ρ is the threading dislocation density, b is the Burger's vector, and l is the dislocation line length in the (001) plane. To get a fully relaxed film, δ should equal to δ_k , and their derivatives with respect to time should also equal to each other, $\partial \delta / \partial t = \partial \delta_k / \partial t$, which gives the following relation:

$$C_f g_f = \rho b v / 2, \text{ (equation 3.4)}$$

where g_f is the film growth rate and v is the dislocation glide velocity. We can rearrange equation 3.4 to get an expression for the threading dislocation density:

$$\rho = 2 C_f g_f / b v. \text{ (equation 3.5)}$$

The threading dislocation density ρ is proportional to the growth gradient so that a higher growth gradient should result in a higher equilibrium threading dislocation density.

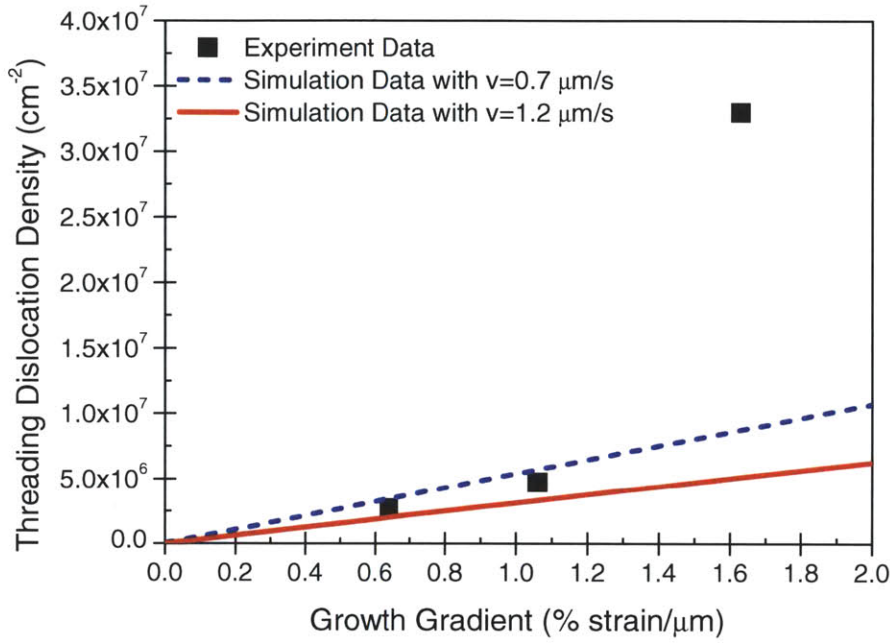


Figure 3.8 Threading dislocation density dependence of growth gradient for (575°C)GaAs_{1-z}Sb_z(z=0-0.49) graded buffer on 6° offcut toward <111>A GaAs (001) substrate.

Using equation 3.5, we calculated and plotted the equilibrium threading dislocation density assuming dislocation glide velocities v of 0.7 $\mu\text{m/s}$ and 1.2 $\mu\text{m/s}$. In our simulation, the growth rate is set as 7.5 $\text{\AA/s}$ based on the actual growth conditions; the in-plane Burgers vector b (i.e., the component of the Burgers vector in the (001) plane) for 60° dislocations is 0.707 a , where a is the lattice constant of the film (here, we take the GaAs lattice constant 5.653 $\text{\AA}$ as a). Based on the simulation, the threading dislocation density should be between the red solid line and blue dash line if the dislocation glide velocities are in the range of 0.7-1.2 $\mu\text{m/s}$. If the dislocation nucleation, not the

dislocation glide, is the dominating process with regard to the increase in threading dislocation density, the expression in equation 3.5 will underestimate the threading dislocation density. Also, if the threading dislocation density is high enough to enable dislocation interactions and multiplication [3.18], the threading dislocation density will deviate from what is expected from equation 3.5, as shown in Figure 3.8 for the large growth gradient.

Comparing the experiment data and the simulation results, we see that for the samples with growth gradients of 0.64% strain/ μm and 1.06% strain/ μm the threading dislocation density is proportional to the growth gradient and the measured threading dislocation density is in keeping with equilibrium theory, assuming reasonable dislocation glide velocities of 0.7-1.2 $\mu\text{m/s}$. However, the threading dislocation density for the sample with a growth gradient of 1.63% strain/ μm deviates substantially from our prediction. The measured threading dislocation density corresponds to a very high (nonequilibrium) level of dislocations, which indicates that dislocation nucleation, interaction, and multiplication occur to high enough levels to allow the threading dislocation density to deviate from our model.

3.5 Conclusion

In this work, we have created a relaxed InP lattice on (001) GaAs substrates with 6° offcut to the $\langle 111 \rangle_A$, while maintaining a low threading dislocation density ($\sim 10^6 \text{ cm}^{-2}$) and a surface morphology suitable for device processing.

Two compositionally graded buffer designs were investigated to enable the InP lattice constant on a (001) GaAs substrate with a 6° offcut toward the <111>A. The first approach consisted of tandem graded layers of $\text{In}_x\text{Ga}_{1-x}\text{As}$ - $\text{In}_y\text{Ga}_{1-y}\text{P}$ with compositional grading of the In concentration to achieve InP on 6° offcut GaAs with a threading dislocation density of $7.9 \times 10^6 \text{ cm}^{-2}$ and a surface roughness of 33.0 nm, with a total thickness of less than 5 μm . The second approach used graded $\text{GaAs}_{1-z}\text{Sb}_z$ alloys and we demonstrated a $\text{GaAs}_{0.51}\text{Sb}_{0.49}$ alloy (with a growth gradient of 1.06% strain/ μm) lattice-matched to InP on 6° offcut bulk GaAs with a threading dislocation density of $4.7 \times 10^6 \text{ cm}^{-2}$ and a roughness of 7.4 nm for a less than 5 μm buffer. The threading dislocation density of the GaAsSb graded buffer can be further lowered to $2.7 \times 10^6 \text{ cm}^{-2}$ if a lower growth gradient (0.64% strain/ μm) is implemented. The modeling of the $\text{GaAs}_{1-z}\text{Sb}_z$ graded layer system showed agreement to dislocation propagation theory up to the point of excessive growth gradient where the results deviated from theory. For the very high growth gradient case, dislocation interactions and multiplication dominate, which gives a very high dislocation density.

InP lattice on offcut GaAs is one key step for the integration of InP on Si. By combining this work with the GaAs on GOI (Ge on insulator) and wafer transfer technology, one should be able to get the SOLES (Si on lattice engineered substrate) wafer with InP as the III-V template layer.

Chapter 4 Channel Electron Mobility of $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ Quantum-Well MOSFET Structures

4.1 Introduction

The superior electron mobility of $\text{In}_x\text{Ga}_{1-x}\text{As}$ alloys has made them a great candidate for high-frequency electronics. $\text{In}_x\text{Ga}_{1-x}\text{As}$ HEMTs incorporate an undoped quantum-well channel and use Schottky gate contacts to modulate the charge density within the channel. However, the Schottky gate contact leads to large gate leakage currents and prevents the large scale integration of these transistors. On the other hand, $\text{In}_x\text{Ga}_{1-x}\text{As}$ MOSFETs can exhibit very low gate leakage because of the gate dielectric, but the need for doping in the channel and the direct contact between the channel and the gate dielectric leads to carrier scattering that reduces carrier mobility and overall device speed.

M. Radosavljevic *et al.* established an integrated high- κ gate stack with an $\text{In}_x\text{Ga}_{1-x}\text{As}$ HEMT structure and demonstrated a high performance $\text{In}_x\text{Ga}_{1-x}\text{As}$ quantum-well MOSFET [4.1]. P. Nagaiah *et al.* reported on their analysis of the channel mobility in $\text{In}_{0.77}\text{Ga}_{0.23}\text{As}$ quantum-well MOSFET structures [4.2]. For such a structure, the gate leakage current is small because of the gate dielectric and the channel; the carrier mobility is high because of the undoped channel and the barrier layers between the channel and the gate dielectric. While the barrier layer(s) is (are) necessary to maintain a high carrier mobility in the channel, a barrier layer also increases the effective oxide thickness which is undesirable.

In this thesis, we present experimental and modeling results on how the

barrier thickness and gate dielectric/barrier layer interfacial defect density affect the carrier mobility in $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ quantum-well MOSFET structures. The modeling results showed good agreement with the experiment results and can be generally used in quantum-well MOSFET design.

4.2 Experiment Procedures

The $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ quantum-well MOSFET structures were deposited on semi-insulating InP (001) substrates in a Thomas Swan/AIXTRON low pressure metal-organic chemical vapor deposition (MOCVD) system with a close-coupled showerhead design. The III-V compound epitaxy was conducted in the CVD mode and Al_2O_3 was deposited *in-situ* in either the ALD mode [4.3] or the CVD mode [4.4]. Trimethylgallium (TMGa), trimethylaluminum (TMAI), trimethylindium (TMIn), arsine (AsH_3) and phosphine (PH_3) were used as the precursors for the III-V compound epitaxy, disilane (Si_2H_6) was used for the n-type modulated doping, and TMAI and isopropyl alcohol (IPA) were used for the deposition of Al_2O_3 . High-purity N_2 was used for both the carrier gas and the purge gas.

III-V compound epitaxy was conducted with a susceptor temperature of 650°C with a chamber pressure of 100 Torr. After the III-V compound epitaxy sequence, under an AsH_3 overpressure, the temperature was ramped down to a growth temperature of Al_2O_3 , which was 370 °C for ALD and 500 °C for CVD, and a chamber pressure was ramped down to 50 Torr. For the ALD mode Al_2O_3 deposition, the pulse time for TMAI (with a partial pressure of 0.03 Torr)

and IPA (partial pressure of 0.20 Torr) was kept at 4 sec, and the N₂ purge time in between TMAI and IPA pulse was also kept at 4 sec. For CVD Al₂O₃, TMAI and IPA were flowed simultaneously into the chamber, where the TMAI and IPA partial pressures were kept as 0.01 and 0.15 Torr, respectively. The *in-situ* process Al₂O₃ deposition ensures that the III-V compound surfaces were never exposed to air prior to deposition of the gate dielectric.

The layer thickness was characterized by cross-sectional transmission electron microscopy (XTEM) with a JEOL 2011 microscope operated at 200 kV and the surface morphology was characterized by atomic force microscopy (AFM) with a Veeco Nanoscope IV. The composition of In_{0.53}Ga_{0.47}As and In_{0.52}Al_{0.48}As was determined by x-ray diffraction (XRD) with a Bruker D8 XRD system.

The Hall mobility was measured by either the Van der Paul method or the Hall bar method. Typically, we used a Van der Paul configuration for ungated structures and a Hall bar configuration for gated Hall structures. Based on the sample and contact dimensions, for the Van der Pauw configuration the Hall measurement error is approximated to be 7.5% [4.5] and for the Hall bar configuration the error is around 12.7% [4.6]

To fabricate structures for Hall measurements, we first defined the mesa and then deposited the ohmic contacts. For the gated Hall measurement, a gate was also deposited. To form the ohmic contacts, Al₂O₃ in the contact areas was etched away with a one minute buffered oxide etch, after which

Au(100nm)/Ge(25nm)/Ni(10nm) layers were deposited by e-beam evaporation followed by a 30 sec 450 °C rapid thermal annealing. To form the gate contacts, 50 nm Al was deposited by thermal evaporator and no thermal annealing was applied afterwards. To avoid any change in the interface quality with time we fabricated the Hall structures right after the deposition of Al_2O_3 on III-V and measured the mobility immediately after the fabrication.

4.3 Optimization of $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ Quantum-Well MOSFET Structures.

To get good quality $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ quantum-well MOSFET structures, we firstly optimized the growth conditions for the $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ quantum-well.

Figure 4.1 (a) shows the AFM of an $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ quantum-well structure grown by MOCVD. In this structure, the InAlAs buffer layer was directly deposited on InP and the structure showed pit-like surface defects. To improve the surface morphology of this $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ quantum well structure, a thin layer of InGaAs (0.3 nm) was deposited in between the InAlAs buffer layer and the InP epi-layer. This dramatically improved the surface morphology as shown in Figure 4.1 (b).

It is well-known that the V/III ratio affects the mobility of the III-V compound semiconductors grown by MOCVD [4.7]. Therefore, in addition to depositing the $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ interlayer to improve the surface morphology, we experimented with the V/III ratio during the growth of the $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ channel layer and found that a V/III ratio of 23 provided for the highest mobility.

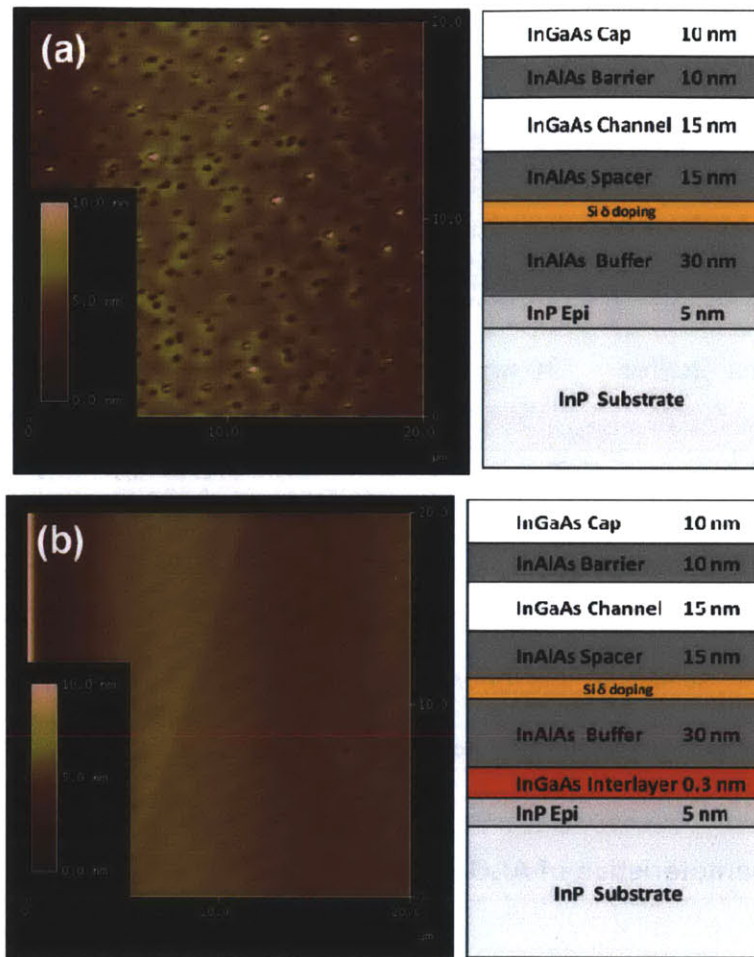


Figure 4.1 AFMs of (a) an $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ quantum-well structure with InAlAs buffer directly on InP, and (b) an $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ quantum-well structure with a thin InGaAs (0.3 nm) interlayer in between the InAlAs buffer and the InP epi-layer.

After having demonstrated good quality $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ quantum-well structures, we deposited Al_2O_3 on these structures by either *in-situ* ALD or *in-situ* CVD. The final surface of the MOSFET structures was very smooth, with a measured RMS surface roughness of 0.2 nm as indicated in Figure 4.2. In this structure, the barrier thickness is zero and Al_2O_3 was deposited by ALD. The

mobility of these structures was then studied and the results will be presented in Section 4.5.

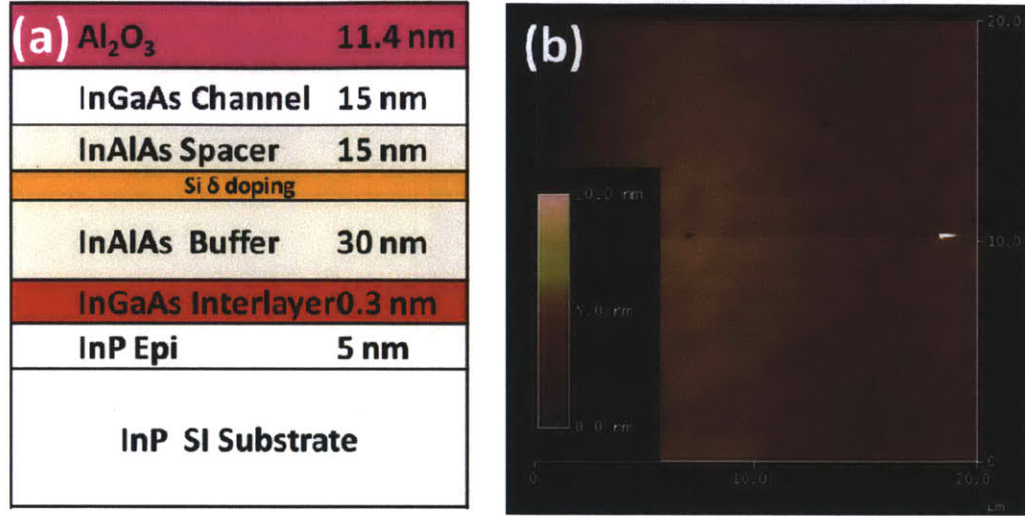


Figure 4.2 (a) Schematic and (b) AFM of an $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ quantum-well MOSFET structure. In this structure, there is no barrier layer and Al_2O_3 was deposited by ALD.

4.4 C-V Characteristics of $\text{Al}_2\text{O}_3/\text{InGaAs}$

C-V characteristics of ALD $\text{Al}_2\text{O}_3/\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ and CVD $\text{Al}_2\text{O}_3/\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ were studied and compared. For $\text{Al}_2\text{O}_3/\text{p-In}_{0.53}\text{Ga}_{0.47}\text{As}$, ALD $\text{Al}_2\text{O}_3/\text{p-In}_{0.53}\text{Ga}_{0.47}\text{As}$ (Figure 4.3 (a)) gives a higher accumulation capacitance and a smaller frequency dispersion at a flat band voltage compared to CVD $\text{Al}_2\text{O}_3/\text{p-In}_{0.53}\text{Ga}_{0.47}\text{As}$ (Figure 4.3 (c)). However, CVD $\text{Al}_2\text{O}_3/\text{p-In}_{0.53}\text{Ga}_{0.47}\text{As}$ shows clear inversion (Figure 4.3 (c)) while ALD $\text{Al}_2\text{O}_3/\text{p-In}_{0.53}\text{Ga}_{0.47}\text{As}$ does not show inversion (Figure 4.3 (a)). This means ALD Al_2O_3 gives a lower interfacial defect density near the valence band edge but a higher defect density near the mid gap and conduction band edge. This conclusion is consistent with the

findings in Al_2O_3 on $n\text{-In}_{0.53}\text{Ga}_{0.47}\text{As}$ samples. Figure 4.3 (b) and (d) give the ALD Al_2O_3 and CVD Al_2O_3 on $n\text{-In}_{0.53}\text{Ga}_{0.47}\text{As}$. It is clear that ALD Al_2O_3 gives a larger frequency dispersion in accumulation, which means that it has a higher interfacial defect density near the conduction band edge. From the C-V characteristics of ALD Al_2O_3 and CVD Al_2O_3 on $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$, we know that these two growth mechanisms give different interfacial defect density.

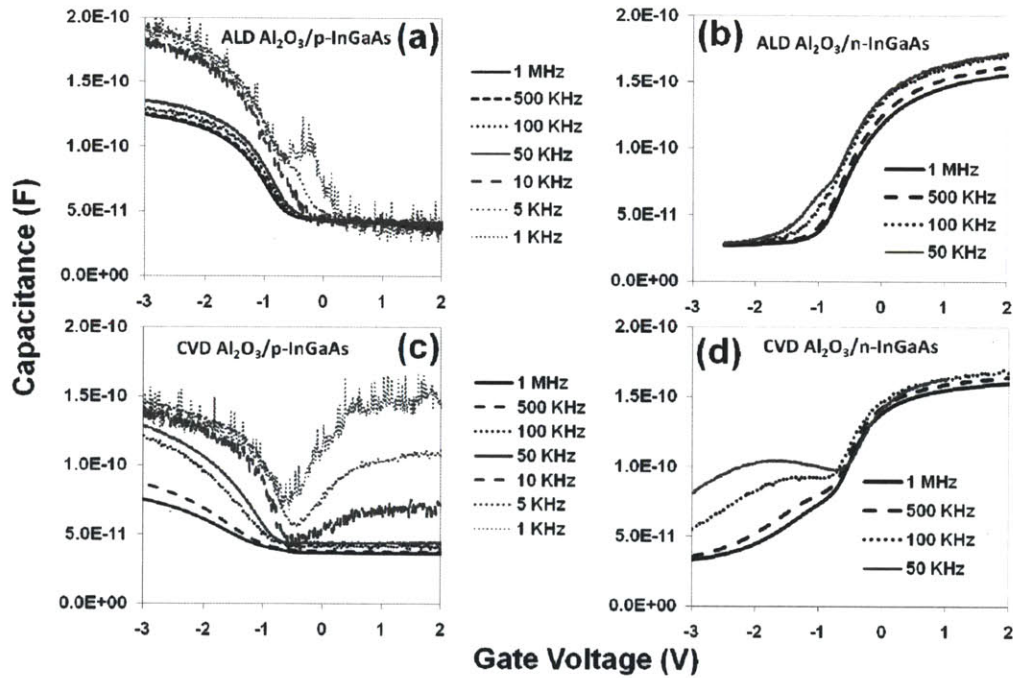


Figure 4.3 C-V characteristics of *in-situ* ALD Al_2O_3 on (a) $p\text{-In}_{0.53}\text{Ga}_{0.47}\text{As}$ and (b) $n\text{-In}_{0.53}\text{Ga}_{0.47}\text{As}$, and *in-situ* CVD Al_2O_3 on (c) $p\text{-In}_{0.53}\text{Ga}_{0.47}\text{As}$ and (d) $n\text{-In}_{0.53}\text{Ga}_{0.47}\text{As}$.

From the mobility point of view, only charged defects will cause coulomb scattering and reduce the channel carrier mobility. Typically the density of charged defects N_{it}^c can be approximated as:

$$N_{it}^c = N_{it}^{d+} + N_{it}^{a-} = \int_{E_F}^{+\infty} D_{it}^d dE + \int_{-\infty}^{E_F} D_{it}^a dE, \text{ (equation 4.1)}$$

where N_{it}^{d+} and N_{it}^{a-} are the density of ionized donor-like and acceptor-like defects at the oxide/semiconductor interface, D_{it}^d and D_{it}^a are the density distribution of donor-like and acceptor-like interfacial traps, and E_F is the Fermi level energy. Therefore, to get the charged defect density N_{it}^c , one need to know D_{it}^d , D_{it}^a and E_F . This requires a better understanding of the types of interfacial traps by techniques like x-ray photoelectron spectroscopy or surface potential fluctuation method [4.8].

There are two important things that I need to point out here. First, the charged interfacial traps are not necessarily located in the band-gap. They can be either in the valence band, or in the band-gap, or in the conduction band. From the conductance-frequency method, we can only get the D_{it} within the band-gap. For the ALD and CVD $\text{Al}_2\text{O}_3/\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ samples in this work, the measured D_{it} within the band-gap is in the range of 10^{12} - $10^{13} \text{ cm}^{-2}\text{eV}^{-1}$. Second, charged interfacial traps can reduce the mobility but do not necessarily shift the flat-band voltage of the MOSFETs [4.9]. This can be explained if there are both negative and positive charges at the gate dielectric/barrier layer interface but the net charge is neutral. In this case, the carriers still get scattered by the local disturbance of these negative and positive charges but the flat-band voltage does not shift due to the neutral net charge [4.9].

4.5 Mobility of $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ Quantum-Well MOSFET Structures

As discussed in Section 1.3.2, there are six major scattering mechanisms that

may affect the carrier mobility of a quantum-well MOSFET: semiconductor (internal) phonon scattering, gate dielectric (remote) phonon scattering [4.10, 4.11], gate dielectric/barrier layer interfacial roughness scattering, coulomb scattering from ionized dopants, coulomb scattering from charges trapped in gate dielectric, and coulomb scattering from gate dielectric/barrier layer interfacial charge.

ALD or CVD Al_2O_3 t_{oxide}	
InGaAs Cap	} t_{barrier}
InAlAs Barrier	
InGaAs Channel	15 nm
InAlAs Spacer	15 nm
Si δ doping	
InAlAs Buffer	30 nm
InGaAs Interlayer	0.3 nm
InP Epi	5 nm
InP SI Substrate	

Figure 4.4 Schematic of quantum-well MOSFET structure fabricated for this study.

In this thesis, we fabricated Al_2O_3 (gate dielectric)/ $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ - $\text{In}_{0.52}\text{Al}_{0.48}\text{As}$ (barrier)/ $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ (channel) structures as shown in Figure 4.4 to study the effect of different scattering mechanisms on the carrier mobility. The structure contains a thin (0.3 nm) $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ interlayer between the InP epi and $\text{In}_{0.52}\text{Al}_{0.48}\text{As}$ buffer, which improved the surface morphology as we found that direct deposition of $\text{In}_{0.52}\text{Al}_{0.48}\text{As}$ on InP introduced a finite density of surface imperfections. The final surface of the MOSFET structures was very

smooth, with a measured RMS surface roughness of 0.2 nm. In addition to the $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ interlayer to improve surface morphology, we experimented with the V/III ratio during the growth of $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ channel layer and determined that a V/III ratio of 23 provided for the highest mobility.

For such a buried channel MOSFET with zero or small gate bias, it is reasonable to assume that the Al_2O_3 phonon scattering [4.12] and interfacial roughness scattering would be negligible. In addition, for our structure the effects of ionized impurity scattering due to the Si δ doping layer could also be considered negligible since the spacer layer was 15 nm thick. Therefore, in this work, we only consider three scattering processes for our experiments and model: semiconductor phonon scattering, coulomb scattering from charges trapped in the Al_2O_3 gate dielectric, and interfacial charge density at the $\text{Al}_2\text{O}_3/\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ interface. Hall measurements were used to examine the mobility and a model was developed to fit the experimental data.

For our experiments, we first changed the Al_2O_3 thickness (t_{oxide}) to determine whether scattering from charges trapped in the oxide is a key factor in determining the channel carrier mobility. We then fixed the oxide thickness and varied the following three factors to study the effect of coulomb scattering from the interfacial charges: the separation between the interfacial charges and the carriers, the density of the interfacial charges, and the density of the carriers. Specifically, we varied the barrier thickness (t_{barrier}) to change the separation between the interfacial charges and the carriers, varied the Al_2O_3 deposition methods to change the density of the interfacial charges, applied a

gate voltage to change the carrier density, and then studied the mobility versus t_{barrier} , the mobility versus different Al_2O_3 deposition methods, and the mobility versus gate voltage.

4.5.1 Mobility versus Oxide Thickness

Figure 4.5 presents the measured sheet carrier density n and electron mobility μ of the fabricated $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ quantum-well MOSFET structures with varying oxide thickness. The carrier density stayed invariant with varying oxide thickness (Figure 4.5 (a)). Changing oxide thickness changes the amount of accumulated oxide charges, which will change the band bending if the charge amount is huge and in consequence change the carrier density. Therefore, invariant carrier density with varying oxide thickness suggests that the amount of charges within the oxide is not significant. This is consistent with the conclusion from the following mobility data (Figure 4.5 (b)).

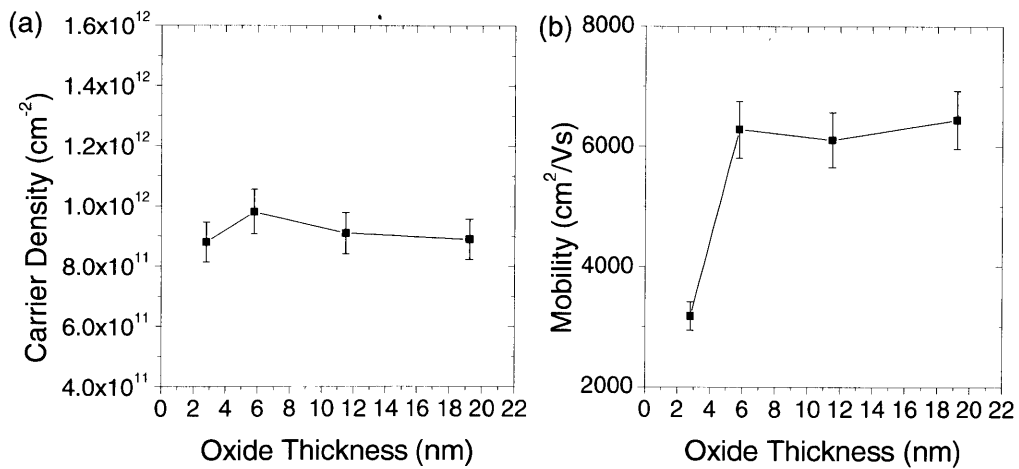


Figure 4.5 (a) Sheet carrier density, and (b) electron mobility of $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ channel versus ALD Al_2O_3 thickness.

The electron mobility was lowest when the oxide thickness was 2.8 nm, but was largely invariant for oxide thicknesses greater than 5.8 nm (Figure 4.5(b)). The low carrier mobility for the structure with a 2.8 nm oxide thickness may be due to two reasons: the top surface of ALD Al_2O_3 may have trapped charges that essentially increase the propensity for coulomb scattering; the thinner oxide makes the ALD $\text{Al}_2\text{O}_3/\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ more susceptible to oxygen or water vapor increasing the possibility for the interface to degrade and become more defective over time. The invariant carrier mobility with increasing oxide thickness for oxide thickness greater than 5.8 nm suggests that scattering from charges within the oxide is not a dominant factor for carrier mobility since if the oxide charges were a problem one would expect that the thicker oxide layer would have a lower mobility since there would be greater cumulative charges. The invariant carrier mobility with varying oxide thickness also suggests that the scattering from the oxide phonons is not a key factor to affect the mobility, which is consistent with reference [4.12] that Al_2O_3 phonons have a minor effect on the mobility even for surface channel device at high electric field.

4.5.2 Mobility Model

As demonstrated above, coulomb scattering from the oxide charge is negligible. Therefore, only two scattering processes are left: semiconductor phonon scattering and coulomb scattering from interfacial charges. In this work, we developed a mobility model which includes the above two scattering

mechanism to fit our experimental data. In the modeling, we assumed the conditions to be at room temperature, 300K. The interfacial defects were assumed to be precisely at the $\text{Al}_2\text{O}_3/\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ interface with a two-dimensional (2D) density of N_{it}^c . At zero gate voltage, the carriers were assumed to be in the center of the channel with a 2D density of n , which can be obtained from Hall measurements. Then the distance d between carriers and interfacial defects becomes: $d = t_{barrier} + \frac{t_{channel}}{2}$, where $t_{barrier}$ is the barrier thickness and $t_{channel}$ is the channel thickness ($t_{channel} = 15 \text{ nm}$ in our case).

The carrier mobility (μ) can be determined using the Matthiessen's rule:

$$\frac{1}{\mu} = \frac{1}{\mu_{phonon}} + \frac{1}{\mu_{coulomb}}, \text{ (equation 4.1)}$$

where μ_{phonon} is the mobility which is limited by semiconductor phonon scattering ($\sim 12000 \text{ cm}^2/\text{Vs}$ at 300K [4.13]) and $\mu_{coulomb}$ is the mobility limited by coulomb scattering from interfacial charges. We calculated $\mu_{coulomb}$ in the following steps based on established methods [4.14-4.17].

$\mu_{coulomb}$ correlates with the transport relaxation time in the following form:

$$\mu_{coulomb} = \frac{e\langle\tau\rangle}{m^*}, \text{ (equation 4.2)}$$

where m^* is the effective mass of the electrons and $\langle\tau\rangle$ is the mean transport relaxation time. $\langle\tau\rangle$ can be obtained from:

$$\langle\tau\rangle = \frac{\int dE \tau(E) E \left(-\frac{\partial f}{\partial E}\right)}{\int dE E \left(-\frac{\partial f}{\partial E}\right)}, \text{ (equation 4.3)}$$

where f is the room temperature Fermi-Dirac distribution function, E is the

electron energy, and $\tau(E)$ is the transport relaxation time. Using the Born approximation, the transport relaxation time $\tau(E)$ is given by:

$$\frac{1}{\tau(E)} = \frac{2\pi}{\hbar} \sum_{\mathbf{k}'} \int_{-\infty}^{\infty} N(z) |u(\mathbf{k} - \mathbf{k}'; z)|^2 (1 - \cos(\theta_{\mathbf{k}\mathbf{k}'})) \delta(E_{\mathbf{k}} - E_{\mathbf{k}'}) dz, \quad (\text{equation 4.4})$$

where $N(z) = N_{it}^c \delta(z - d)$ with N_{it}^c as the interfacial charged defect density and d as the distance between the carriers and the interfacial defects, $u(\mathbf{k} - \mathbf{k}'; z)$ is the screened scattering potential dependent on the separation d between interfacial defects and carriers as well as on the carrier density n , the angle $\theta_{\mathbf{k}\mathbf{k}'}$ is the scattering angle, and $E_{\mathbf{k}}$ is the 2D carrier energy for wave vector $\mathbf{k}$. $u(\mathbf{k} - \mathbf{k}'; z)$ can be calculated within the random phase approximation, which is detailed in reference [4.17].

According to the above model, coulomb scattering depends on the separation d between interfacial defects and carriers, the density of charged interfacial traps N_{it}^c , and the carrier density n . We conducted experiments to study the carrier mobility by varying d , N_{it}^c and n . We also conducted modeling to fit the experiment data.

4.5.3 Mobility versus Barrier Thickness

By varying the barrier thickness t_{barrier} , one changes the separation d between carriers and interfacial defects ($d = t_{\text{barrier}} + \frac{t_{\text{channel}}}{2}$). Figure 4.6 shows the experimental and modeling results of the channel mobility versus $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}/\text{In}_{0.52}\text{Al}_{0.48}\text{As}$ barrier thickness t_{barrier} . The measured channel

electron mobility increases from 2378 cm²/Vs to 10312 cm²/Vs as the barrier thickness increases from 0 to 16 nm. At the beginning the carrier mobility increases rapidly with increasing barrier thickness, but starts to plateau for the thickest barriers.

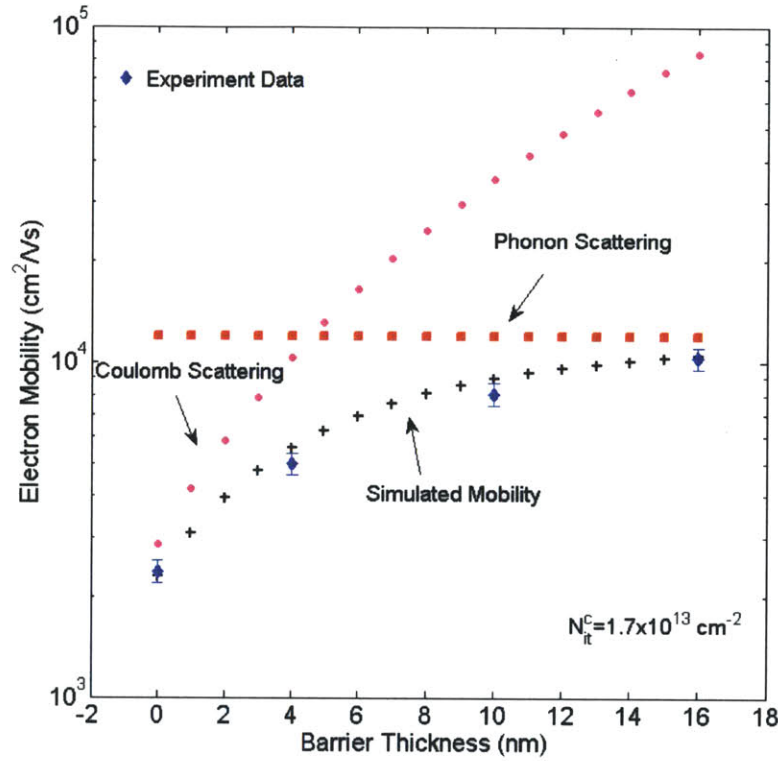


Figure 4.6 Channel electron mobility versus In_{0.53}Ga_{0.47}As/In_{0.52}Al_{0.48}As barrier thickness. The blue “♦” symbols indicate the measured mobility. The magenta “•” symbols indicate the mobility limited by coulomb scattering, the red “■” symbols indicate the mobility limited by phonon scattering, and the black “+” symbols indicate the simulated final mobility. In the modeling, n was set as $1.0 \times 10^{12} \text{ cm}^{-2}$ based on the Hall measurement data and N_{it}^c was used as the only variable to fit the experimental data points.

Our mobility model predicts this effect as when the barrier thickness is less than 5 nm, the mobility is mainly dominated by coulomb scattering, and the relationship to barrier thickness is sub-exponential but super-linear; when the barrier thickness is larger than 5 nm, the mobility is mainly dominated by phonon scattering, which leads to the saturation in mobility for large barrier thickness. In the modeling, n was set as $1.0 \times 10^{12} \text{ cm}^{-2}$ based on the Hall measurement data and N_{it}^c was used as the only variable to fit the experimental data points. By fitting the simulated mobility with the experimental data, we were able to extract an interfacial charged defect density N_{it}^c of $1.7 \times 10^{13} \text{ cm}^{-2}$ for the ALD $\text{Al}_2\text{O}_3/\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ interface.

4.5.4 Mobility versus Defect Density

According to our modeling, at a barrier thickness of 4 nm, the coulomb scattering should dominate. However, that was with our assumed interfacial defect density N_{it}^c of $1.7 \times 10^{13} \text{ cm}^{-2}$ and carrier density n of $1.0 \times 10^{12} \text{ cm}^{-2}$. We conducted another experiment to better understand the correlation between interfacial defect density and carrier mobility. We varied the interfacial defect density by replacing the ALD Al_2O_3 with CVD Al_2O_3 , which would change the interface and corresponding interfacial defect density as discussed in section 4.4.

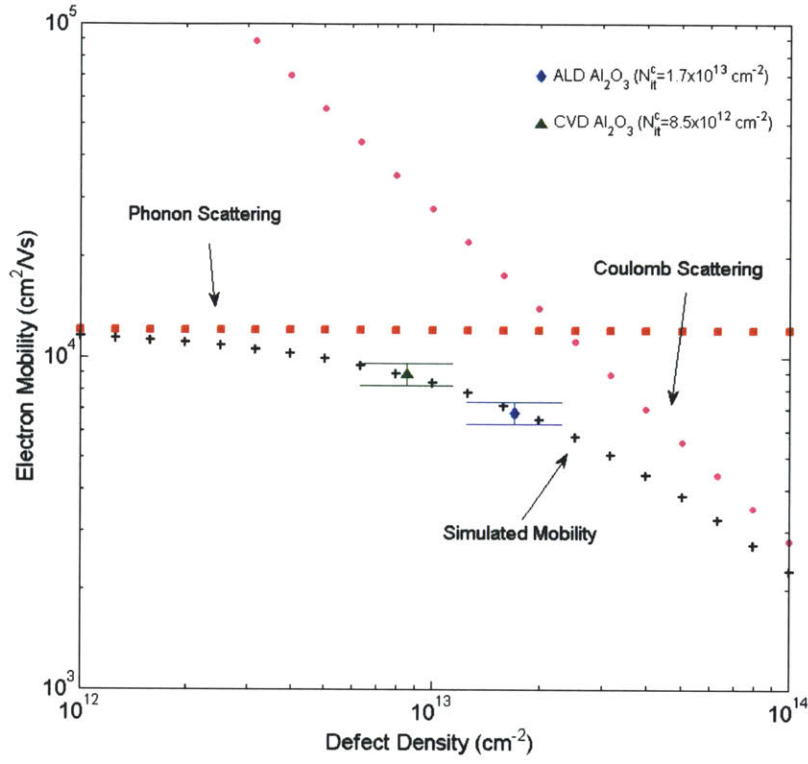


Figure 4.7 Channel electron mobility versus the charged defect density. The blue “♦” symbol and the green “▲” symbol indicate the measured mobility for structure with CVD $\text{Al}_2\text{O}_3/\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ interface and ALD $\text{Al}_2\text{O}_3/\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ interface, respectively. The magenta “●” symbols indicate the mobility limited by coulomb scattering, the red “■” symbols indicate the mobility limited by phonon scattering, and the black “+” symbols indicate the simulated final mobility. In the modeling, t_{barrier} was set as 4 nm; n was set as $1.5 \times 10^{12} \text{ cm}^{-2}$ based on the Hall measurement data; and N_{it}^c was extracted by fitting the modeling data.

Figure 4.7 compares the carrier mobility for structures with CVD Al_2O_3 and ALD Al_2O_3 . Application of a CVD Al_2O_3 gate oxide with our structure improved the carrier mobility to our maximum measured mobility of $8883 \text{ cm}^2/\text{Vs}$, which

was much higher than the $6807 \text{ cm}^2/\text{Vs}$ mobility measured for the corresponding structure with an ALD Al_2O_3 oxide. By fitting the data with our model we were able to determine the charged interfacial defect density (see Figure 4.7). In the modeling, the barrier thickness was set as 4 nm, which gave an 11.5 nm separation between carriers and interfacial defects ($d = t_{\text{barrier}} + \frac{t_{\text{channel}}}{2}$); the carrier density n was set as $1.5 \times 10^{12} \text{ cm}^{-2}$ based on the Hall measurement data. The calculated charged defect density for the CVD $\text{Al}_2\text{O}_3/\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ interface was $8.5 \times 10^{12} \text{ cm}^{-2}$, which is much lower than that calculated for the ALD $\text{Al}_2\text{O}_3/\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ interface, $1.7 \times 10^{13} \text{ cm}^{-2}$. Figure 4 also show that with a barrier thickness of 4 nm and a carrier density of $1.5 \times 10^{12} \text{ cm}^{-2}$, an interfacial charged defect density lower than $2.3 \times 10^{13} \text{ cm}^{-2}$ is required to render the effects of coulomb scattering inconsequential as compared to phonon scattering effects.

Another method to change the interfacial defect density is to change the barrier layer. Compared to InAlAs, InP is more immune to oxidation. Therefore, InP is a better candidate for the barrier layer for the MOSFETs with a gate recess process. In this work, the mobility for InGaAs quantum-well MOSFET structures with InP and InGaAs/InAlAs barrier layers was compared.

Figure 4.8 gives the mobility of the MOSFET structures with an InGaAs/InAlAs barrier and an InP barrier. For the InGaAs/InAlAs barrier case, the high-k/III-V interface is ALD $\text{Al}_2\text{O}_3/\text{InGaAs}$ while for the InP barrier case, the high-k/III-V interface is ALD $\text{Al}_2\text{O}_3/\text{InP}$. As shown in Figure 4.8, the MOSFET structure with

InP barrier gave a much lower carrier mobility than that with an InGaAs/InAlAs barrier. This further demonstrated that the carrier mobility for quantum-well MOSFET structures strongly depends on the properties of a high-k/III-V interface. The result also suggested that the ALD $\text{Al}_2\text{O}_3/\text{InP}$ interface must have higher interfacial charges than the ALD $\text{Al}_2\text{O}_3/\text{InGaAs}$ interface. Here, the carrier density for the MOSFET structure with an InGaAs/InAlAs barrier was very close to that of the InP barrier case.

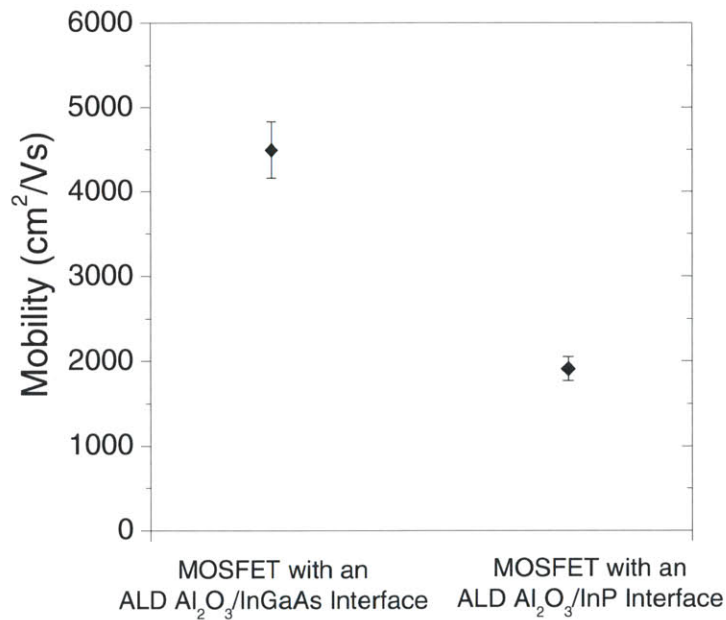


Figure 4.8 Comparison of the channel electron mobility for MOSFET structures with an ALD $\text{Al}_2\text{O}_3/\text{InGaAs}$ interface and an ALD $\text{Al}_2\text{O}_3/\text{InP}$ interface

4.5.5 Mobility versus Carrier Density

According to the mobility model discussed in section 4.4.2, coulomb scattering

is dependent on the carrier density, because carriers significantly screen the electric field of the interfacial charges and therefore change the effective potential in the channel. In our experiment, we also noticed that for two samples with the same barriers (4 nm InGaAs/InAlAs) and the same oxides (ALD Al₂O₃) but a different carrier density, the sample with a carrier density of $1.5 \times 10^{12} \text{ cm}^{-2}$ showed a higher mobility than the sample with a carrier density of $1.0 \times 10^{12} \text{ cm}^{-2}$. This suggested that given other conditions being the same, the higher carrier density sample will have a higher carrier mobility. This can be explained by the coulomb scattering model. Higher carrier density provides a larger screening factor for the potential and therefore reduces the effect of coulomb scattering.

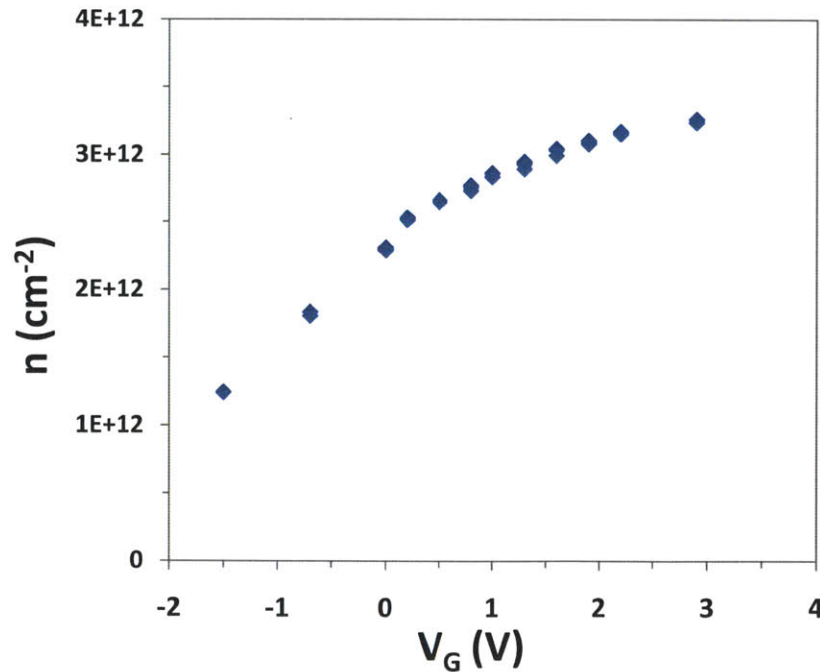


Figure 4.9 Channel carrier density versus gate voltage.

We intentionally changed the carrier density by applying a gate voltage and we studied the effect of carrier density on the mobility. As shown in Figure 4.9, by applying gate voltages from -1.5 V to 3 V, we were able to tune the carrier density from $1.2 \times 10^{12} \text{ cm}^{-2}$ to $3.2 \times 10^{12} \text{ cm}^{-2}$. Figure 4.10 plots the mobility versus the carrier density for an $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ MOSFET structure with the CVD Al_2O_3 and a 4 nm InGaAs/InAlAs barrier. It is clear that when the carrier density $< 2.7 \times 10^{12} \text{ cm}^{-2}$, the mobility increases with increasing carrier density; when the carrier density $> 2.7 \times 10^{12} \text{ cm}^{-2}$, the mobility starts to decrease with increasing carrier density.

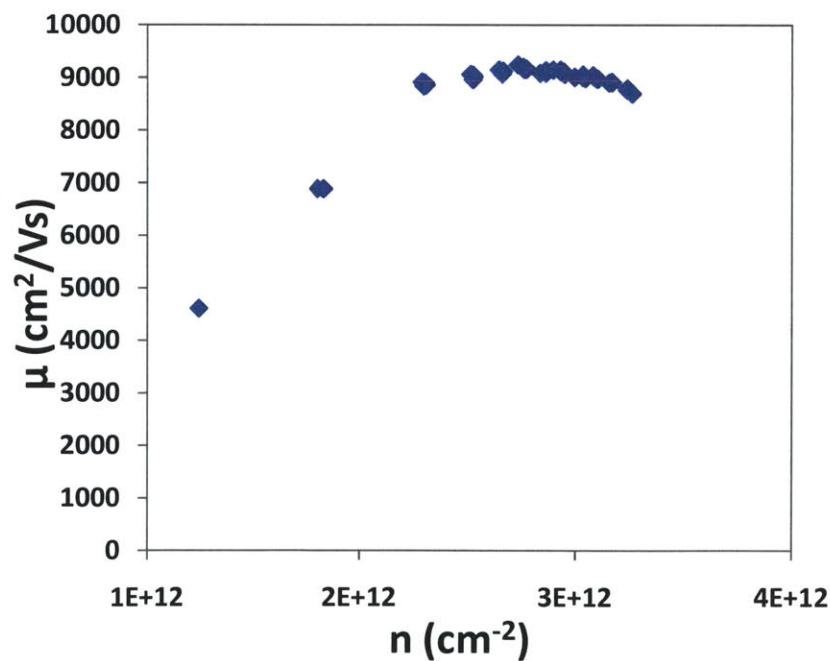


Figure 4.10 Channel electron mobility versus carrier density.

The phenomena can be explained by the coulomb scattering discussed in

section 4.4.2. Applying a gate voltage changes the carrier density as shown in Figure 4.9, while in the mean time, the gate voltage also shifts the position of the carriers, which leads to a change of the separation between the interfacial charges and the carriers. Sweeping the gate voltage from -1.5 V to 3 V, when the carrier density is less than $2.7 \times 10^{12} \text{ cm}^{-2}$, the effect of the increase of carrier density with increasing gate voltage dominates and the mobility increases; when the carrier density is larger than $2.7 \times 10^{12} \text{ cm}^{-2}$, the carrier density starts to saturate with increasing gate voltage and the effect of the decrease of the separation between the interfacial charges and the carriers dominates, which leads to the decrease of the mobility.

4.6 Conclusion

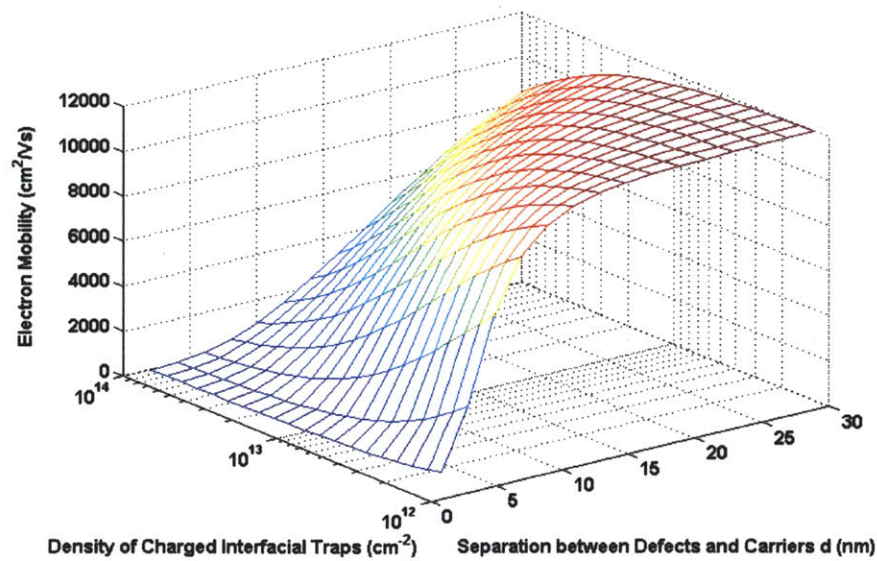


Figure 4.11 Channel electron mobility versus density of charged interfacial traps and the separation between interfacial traps and carriers for a MOSFET structure with a carrier density of 10^{12} cm^{-2} .

We studied the channel electron mobility of $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ quantum-well MOSFET structures. First, we experimentally demonstrated that the effect of the charges within the gate dielectric (Al_2O_3) is negligible. We then studied the effects of the barrier thickness, the high-k/III-V interface properties, and the carrier density on the carrier mobility. Typically, the mobility increases with increasing barrier thickness, decreasing interfacial charge density and increasing carrier density. A mobility model was developed to quantify the effect of these three factors on the mobility. For an $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ quantum-well MOSFET structure with a carrier density of 10^{12} cm^{-2} , the mobility versus density of charged interfacial traps and the separation between interfacial charges and carriers is given in Figure 4.11. This model can be used to sample MOSFET designs that are limited by phonon scattering and gate dielectric/barrier layer interfacial charge density.

Chapter 5 Summary, Suggestions for Future Work

5.1 Summary of Results

This thesis work is motivated by the desire to realize high quality III-V/Si MOSFETs. Two key issues in the III-V/Si MOSFETs were studied. The first one is to integrate InP lattice on Si substrates and the second one is to study InP-based InGaAs MOSFETs.

Specifically, to engineer the lattice constant from Si to that of InP, we first demonstrated GaAs on Si by taking a 6° offcut GOI wafer as the growth template and depositing GaAs on the GOI template; then we established the InP lattice constant on the 6° offcut GaAs by using MOCVD grown compositionally-graded InGaAs-InGaP alloys or GaAsSb alloys. In this thesis, we mainly focused on establishing the InP lattice constant on 6° offcut GaAs. For the tandem graded InGaAs-InGaP alloys with a compositional grading of the In concentration, an $\text{In}_x\text{Ga}_{1-x}\text{As}$ graded buffer was grown at 700 °C for low In concentration ($X_{\text{In}}=0-0.10$); then the growth temperature was decreased to 450°C for high In concentration ($X_{\text{In}}=0.10-0.30$) to suppress phase separation by reducing the In surface diffusivity; the growth temperature was then increased to 650°C and the graded $\text{In}_y\text{Ga}_{1-y}\text{P}$ system was implemented to continue grading the lattice constant from $\text{In}_{0.30}\text{Ga}_{0.70}\text{As}$ to InP. This tandem approach was found to be necessary because phase separation in the InGaAs alloys leads to surface roughening and a high threading dislocation density when grading lattice constants to be greater than that of $\text{In}_{0.30}\text{Ga}_{0.70}\text{As}$. This approach allowed us to achieve InP on 6° offcut GaAs with a threading dislocation density of $7.9 \times 10^6 \text{ cm}^{-2}$ and an RMS surface roughness of 30.0 nm

on a 40 μm x 40 μm scale. For the graded GaAsSb alloys with compositional grading of the Sb concentration, due to the low surface diffusivity and surfactant nature of Sb, the mixed-anion GaAsSb alloys grown at 575°C did not exhibit phase separation, resulting in high quality InP lattice constant films on GaAs without the need to transition to another material system for compositional grading. We demonstrated a GaAsSb alloy on GaAs (with a growth gradient of 1.06% strain/ μm) lattice-matched to InP with a threading dislocation density of $4.7 \times 10^6 \text{ cm}^{-2}$ and a RMS roughness of 7.4 nm on a 40 μm x 40 μm scale. It was further demonstrated that the threading dislocation density of the GaAsSb graded buffer can be lowered to $2.7 \times 10^6 \text{ cm}^{-2}$ by using a slower growth gradient (0.64% strain/ μm).

For the InP-based InGaAs MOSFETs, we mainly studied the mobility of $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ quantum-well MOSFETs by using Hall measurements. Quantum-well MOSFET structures have a high channel carrier mobility due to the undoped channel and the barrier layers between the channel and the gate dielectric. In this thesis, we experimentally examined the effects on the channel carrier mobility of trapped charges in high-k dielectric, barrier layer thickness, charged interfacial defect density, and carrier density. It was found that the effect of the charges within the gate dielectric (Al_2O_3) is negligible while the mobility increases with increasing barrier thickness, decreasing interfacial charge density and increasing carrier density. For an $\text{In}_{0.53}\text{Ga}_{0.47}\text{As}$ MOSFET structure with a 4 nm InGaAs/InAlAs barrier, a 11.4 nm CVD Al_2O_3 and a carrier density of $2.7 \times 10^{12} \text{ cm}^{-2}$, a mobility of 9243 cm^2/Vs was demonstrated. A mobility model was developed to quantify the effects of the

barrier layer thickness, charged interfacial defect density, and carrier density on the channel carrier mobility. This model can be used to sample various MOSFET designs that are limited by phonon scattering and gate dielectric/barrier layer interfacial charge density.

5.2 Suggestions for Future Studies

Clearly, we have achieved a lot in the work and have explained many phenomena observed in this thesis, however, there is still more to achieve.

For the integration of InP lattice on Si, we solved one big issue, which is establishing InP lattice on offcut GaAs. We also demonstrated GaAs on Si by using an offcut GOI wafer as the growth template. However, there are still two more subjects that someone can work on to get to the final SOLES structure with InP as the III-V device template (Figure 1.3). The first one is to get the InP lattice on Si by combining InP on offcut GaAs and GaAs on GOI, which will be very straightforward. The second one is to get the final SOLES structure with InP template by using smart cut and wafer bonding techniques.

For the InGaAs MOSFETs, we mainly studied the mobility of quantum-well InGaAs MOSFET structures by using Hall measurements and modeling. In reality, there is a lot of work that needs to be done from the mobility point of view for the III-V MOSFETs, which including characterizing the type of the interfacial defects (eg. donor-like or acceptor-like) within the valence band, band-gap, or conduction band to provide the correlation between the

interfacial defects and the density of charged defects, studying the mobility of III-V MOSFETs by transfer characteristics of a MOSFET device, etc. Researchers can also work on the optimization of III-V MOSFET structures by considering both the mobility and the sub-threshold swing.

By achieving both the integration of InP on Si and high performance III-V MOSFETs, one should be able to obtain high performance III-V/Si MOSFETs.

Appendix: MATLAB Code of the Mobility Model for InGaAs Quantum-Well MOSFETs

```

clear;
close;
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
pi=3.1415926; %pi
e=1.6e-19; % electron charge
m=9.1e-31; % electron mass
kb=1.38e-23; % boltzman constant
h=1.05e-34; % redulced plank constant
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%Parameters%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
em=0.041*m; % effective mass of electron in InGaAs
ki=13.9*8.85e-12; % dielectric constant for InGaAs
ka=9.0*8.85e-12; % dielectric constant for Al2O3
k=(ki+ka)/2; % effective dielectric constant
g=1;          % degeneracy
n=2e12*1e4; % carrier density
T=300; %temperature
stept=0.1; % step size for theta
stepu=2e-21; % step size for miupri

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%intermediate variable%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%
kf=sqrt(2*pi*n/g); % fermi vector
ef=h^2*kf^2/2/em; % fermi energy
stepe=ef/20; % step size for energy
beta=1/(kb*T); % beta corresponding to temperature
nf=g*em/pi/h^2; % density of states at fermi energy
miu=1/beta*log(-1+exp(beta*ef)); % chemical potential
q0=2*pi*e^2*nf/k; % zero temp screening factor
%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%intermediate variable%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%%

for i=1:15;
    d(i)=i*2*1e-9; % separation between defects and carriers
    in unit of [m]

```

```

    dnm(i)=i*2; % separation between defects and carriers in
unit of [nm]
for j=1:20
    count=j
    up(j,i)=1.2e4; % mobility limited by phonon scattering
    nn(j)=1e16*10^(j/10); % charged interfacial trap density
in unit of [m^-2]
    avertt(j,i)=0;
    avertt1(j,i)=0;
    avertt2=0;
for ee=ef/1000:stepe:10*ef;
    kk=sqrt(ee*2*em/h^2);
    revtt(j,i)=0;
    for theta=0.001:stept:2*pi+0.001;
        screening=0;
        q=2*sin(theta/2)*kk;
        for miupri=0:stepu:miu+1e-18;
            kfmui=sqrt(2*em*miupri/h^2);
            if q<=2*kfmui
                inside=beta/4*nf/cosh(beta*(miu-miupri)/2)^2;
                insideint=stepu*inside;
                screening=screening+insideint;
            end
            if q>2*kfmui
                inside=beta/4*nf/cosh(beta*(miu-miupri)/2)^2*(1-
sqrt(1-(2*kfmui/q)^2));
                insideint=stepu*inside;
                screening=screening+insideint;
            end
        end
        end
        qs=screening*2*pi*e^2/k;
        u(i)=(2*pi*e^2).exp(-q*d(i))./(k*(q+qs));
        revt(j,i)=2*pi*em/h/h/h/((2*pi)^2)*nn(j)*(u(i)^2)*(1-
cos(theta))*stept;
        revtt(j,i)=revtt(j,i)+revt(j,i);
    end
    tt(j,i)=1/revtt(j,i);

```

```

    avert1(j,i)=stepe*tt(j,i)*ee*exp((ee-ef)*beta)/(exp((ee-ef)*beta)+1)^2;
    avert2=stepe*ee*exp((ee-ef)*beta)/(exp((ee-ef)*beta)+1)^2;
    avertt1(j,i)=avertt1(j,i)+avert1(j,i);
    avertt2=avertt2+avert2;
    avertt(j,i)=avertt1(j,i)/avertt2;
end
avertt(j,i)=avertt1(j,i)/avertt2;
uc(j,i)=e*avertt(j,i)*1e4/em % mobility limited by coulomb
scattering
ut(j,i)=1/(1/up(j,i)+1/uc(j,i)) % total mobility
nit(j)=nn(j)/1e4 % charged interfacial trap density in unit of
[cm^-2]
end
mesh(dnm,nit,ut) % plot mobility versus separation d and
charged defect density

```

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